



PL-900

POWER PLATFORM

[COURSE SLIDES](#) · [NON-WSQ](#)

PL-900 Microsoft Power Platform Fundamentals Training

Course Code: C1358

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE

Microsoft Power Platform, low-code application development, data analytics and cloud solutions.

DELIVERS

Courses on Power Platform, Power BI, Microsoft Azure and business intelligence.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1

Who you are

Your name, your organisation and your role.

2

Your background

Your experience with Excel, SharePoint, Power BI or any low-code tool — if any.

3

What you want

One manual process in your work you would like to automate after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

Be punctual; return from breaks on time.

6

Full attendance — attend both days to get the most from the course.

Download Your Course Material

1

1 · Go to the LMS portal

Open <https://lms-tms.tertiaryinfotech.com> in your browser.

2

2 · Log in

Sign in with the account e-mail you used to register for this course.

3

3 · Open this course Select 'PL-900 Microsoft Power Platform Fundamentals Training'.

4

4 · Download

Download the Trainer Slides, Learner Guide and Lesson Plan (PDF).

5

5 · Get the labs

Clone the lab repository from GitHub — the link is on the course page.

6

6 · Keep them open

You may refer to these materials throughout the course.

Lesson Plan — 2 Days · 7.5 instructional hours per day

Session	Day 1 — foundations, Dataverse and apps	Day 2 — automation, analytics and agents
Morning (9:30)	Introductions · ground rules · course outline	Day 1 recap and Q&A
Morning block	Topic 1: Microsoft Power Platform (Labs 1–2)	Topic 3 continues (Labs 7–8)
Late morning	Topic 2: Dataverse (Labs 3–5)	Topic 4: Power Automate (Labs 9–11)
Lunch (12:35)	30-minute lunch break	30-minute lunch break
Afternoon	Topic 2 labs continue	Topic 5: Power BI (Labs 12–13)
Late afternoon	Topic 3: Power Apps (Lab 6)	Topic 6: Copilot Studio (Labs 14–15)
Close	Day 1 recap and Q&A	Course recap, exam preparation guidance and feedback

WHEN IT MATTERS Each day runs 9:30am to 5:30pm — 7.5 instructional hours of teaching and hands-on labs, plus a 30-minute lunch break.

Course Outline

1

Topic 1 — Introduction to Microsoft Power Platform

Business value · connectors · DLP, compliance, privacy and accessibility.

2

Topic 2 — Introduction to Dataverse

Tables, columns, relationships · environments · business rules · administration.

3

Topic 3 — Introduction to Power Apps

Canvas apps · model-driven apps · Copilot-assisted app creation.

4

Topic 4 — Introduction to Power Automate

Cloud, desktop and business process flows · triggers, actions, approvals.

5

Topic 5 — Introduction to Power BI

Connect, transform, model, visualise · dashboards and sharing.

6

Topic 6 — Power Virtual Agents / Copilot Studio

Agents, topics, knowledge, tools · publishing and troubleshooting.

The Contoso Case Study — one scenario, fifteen labs

1

Store facility request

Store facility request and room data in a structured, secure way.

2

Let employees submit

Let employees submit requests from a phone and let coordinators manage them on a desktop.

3

Notify the facilities

Notify the facilities team automatically when a high-priority request arrives.

4

Report on request

Report on request volume, cost and turnaround time for management.

5

Answer common facilities

Answer common facilities questions with a self-service agent.

6

Govern the whole

Govern the whole thing with environments, DLP policies and solution-based ALM.

Your Lab Environment

1

Power Apps maker portal

make.powerapps.com — build tables, canvas apps and model-driven apps.

2

Power Automate

make.powerautomate.com — build cloud flows, approvals and desktop flows.

3

Power BI

app.powerbi.com plus Power BI Desktop — model, visualise, publish and share.

4

Copilot Studio

copilotstudio.microsoft.com — build, publish and monitor agents.

5

Power Platform admin center

admin.powerplatform.microsoft.com — environments, DLP and analytics.

6

Lab files on GitHub

Mock data, worksheets and step-by-step instructions for all 15 labs.

TOPIC 01

Introduction to Microsoft Power Platform

Business value · connectors · DLP, compliance, privacy and accessibility

Key Concepts — Introduction to Microsoft Power Platform

1

What Power Platform is

A low-code suite — Power Apps, Power Automate, Power BI, Copilot Studio and Power Pages — sitting on Dataverse and 1,400+ connectors so business users build solutions without writing traditional code.

2

Why it exists

Changing workforce expectations, the rising cost of custom development, and the need to be agile and scale delivery without growing the developer headcount.

3

Middleware role (K1)

Power Platform acts as integration middleware: connectors, Dataverse and flows sit between systems that were never designed to talk to each other.

4

Platform types (K3)

Solutions run across web, mobile, desktop and Teams — the same app and data model surface on every platform without a rewrite.

5

Connectors

Standard, premium and custom connectors expose an API as a set of triggers and actions; a custom connector wraps any REST API you own.

6

Governance from day one

DLP policies, environment strategy, compliance, data privacy and accessibility guardrails must be designed in, not bolted on.

WHY POWER PLATFORM EXISTS

Analyse. Act. Automate.

Power Platform lets the people who understand the business build the solution — without waiting for a development team.

The Five Power Platform Services

1

Power Apps

Build low-code business applications that run in a browser, on mobile and inside Teams.

2

Power Automate

Automate repetitive work with cloud flows, desktop flows (RPA) and business process flows.

3

Power BI

Connect, model, visualise and share business data as reports and dashboards.

4

Copilot Studio

Build conversational agents that answer questions and take action — formerly Power Virtual Agents.

5

Power Pages

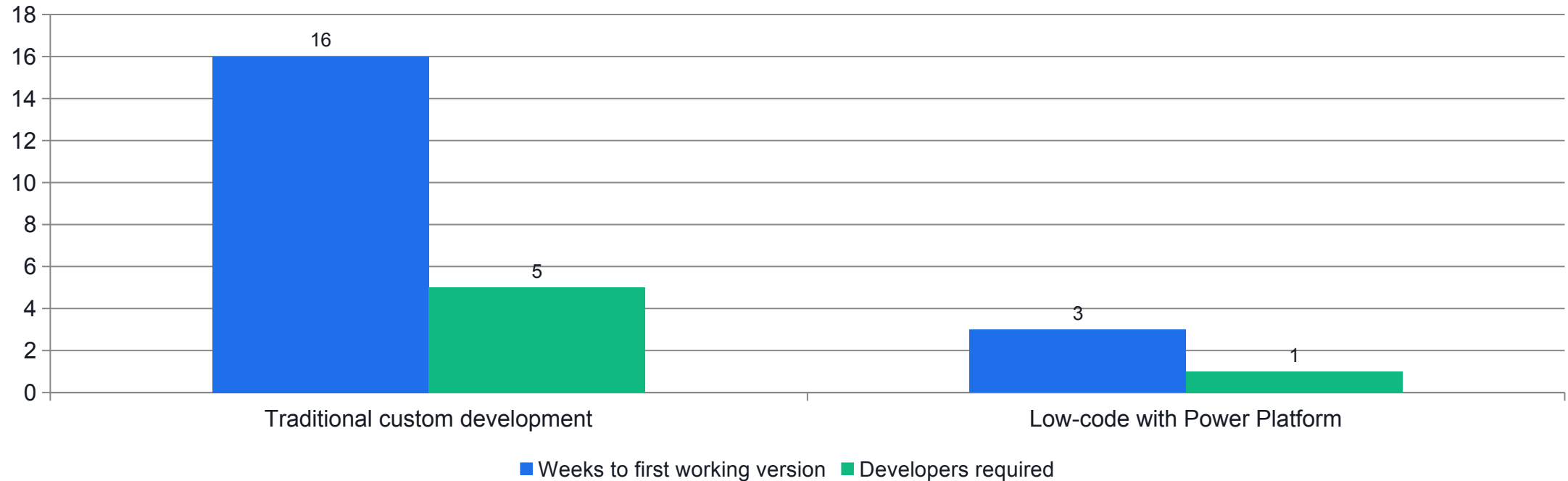
Build secure, data-driven external-facing websites on the same Dataverse data.

6

Dataverse + connectors

The shared data platform and the 1,400+ connectors every service is built on.

The Business Case for Low-Code



WHAT THE DATA SHOWS

Typical low-code delivery reaches a working first version in a fraction of the time and with a fraction of the specialist headcount — which is why the business case is usually speed and capacity, not licence cost.

Four Forces Driving Adoption

1

Changing workforce expectations

Staff expect the same quality of tooling at work that they have on their phones.

2

Rising cost of custom development

Specialist developer time is scarce and expensive; demand outstrips supply.

3

The need to be agile

Processes change faster than a traditional development cycle can deliver.

4

Scaling delivery

Empowering business makers multiplies delivery capacity without multiplying headcount.

Power Platform as Integration Middleware (K1)

Middleware capability	How Power Platform provides it	Contoso example
Connect disparate systems	1,400+ connectors expose each system as triggers and actions	Outlook + SharePoint + Dataverse
Translate data between systems	Dataverse as the common schema; Power Query and expressions transform	Email text becomes a typed request row
Orchestrate a process	Cloud flows sequence the work across systems, with conditions and loops	Request → approval → notify → update
Expose and consume APIs	Custom connectors wrap any REST API; connectors call APIs for you	Legacy Building Services REST API
Enforce policy in transit	DLP policies control which connectors may be combined	Business vs non-business groups

WHEN IT MATTERS This is why Power Platform answers the 'Applications Integration' C1358: it is the middleware layer between systems that were never designed to talk to each other.

Connector Types — and what they cost you (K5)

Type	What it is	Examples	Licence
Standard	Included with most Microsoft 365 plans	Outlook, SharePoint, Teams, OneDrive, Excel	Included
Premium	Requires a Power Apps / Power Automate licence	Dataverse, SQL Server, Salesforce, HTTP	Per user or per app
Custom	You build it to wrap your own REST API	Contoso Building Services API	Premium
On-premises	Reaches a server inside your network via a gateway	On-prem SQL Server, file share	Premium + gateway

WHEN IT MATTERS A connector is an API wrapper: its triggers are the events you can subscribe to and its actions are the operations you can call.

Creating a Custom Connector



WHY IT MATTERS

A custom connector turns any REST API your organisation owns into a first-class Power Platform building block that makers can use without writing code.

Power Platform and Microsoft 365 (K3)

1

Outlook

Trigger on mail arriving; send notifications and actionable approvals.

2

SharePoint

Use lists as a data source; store and route documents.

3

Teams

Embed apps as tabs, deliver approvals, and post adaptive cards.

4

Excel

Import data to start an app; read and write rows in a workbook.

5

Office 365 Users

Look up the signed-in user, their manager and their profile.

6

Dynamics 365

Model-driven apps and Dynamics share the same Dataverse platform.

Governance from Day One

1

Data loss prevention (DLP)

Policies group connectors as Business, Non-business or Blocked — and forbid mixing groups.

2

Environments

Separate dev, test and production so experiments never touch live data.

3

Compliance

Power Platform inherits Microsoft 365 compliance, auditing and data residency.

4

Data privacy

Personal data is governed by tenant policy, retention and the security model.

5

Accessibility

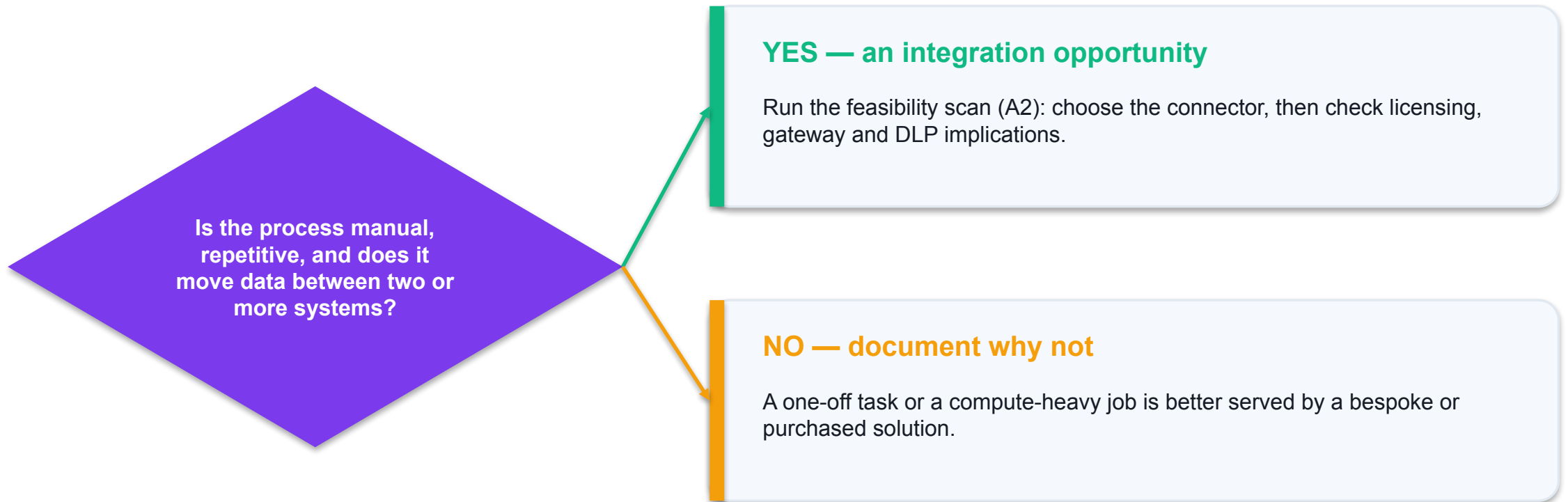
The Accessibility Checker in Power Apps flags contrast, labels and tab order issues.

6

Monitoring

The admin center reports environment usage, flow runs and app adoption.

Is this a Power Platform opportunity? (A1)



Recording the rejected options is part of a defensible feasibility assessment — not every process should become an app.

Validate the Lab Environment and Tour the Maker Portals

LAB 1

You will be able to sign in to the Power Platform, confirm the environment and licence, and navigate the four maker portals used throughout the course.

Before building anything you confirm you have a working environment. You sign in to the Power Apps maker portal, verify which environment you are in, check your licence, and tour the Power Automate, Power BI and Copilot Studio portals so you know where each service lives.

YOU'LL BUILD

A validated Power Platform environment plus a portal reference sheet recording your environment name, region, licence type and the Dataverse state.

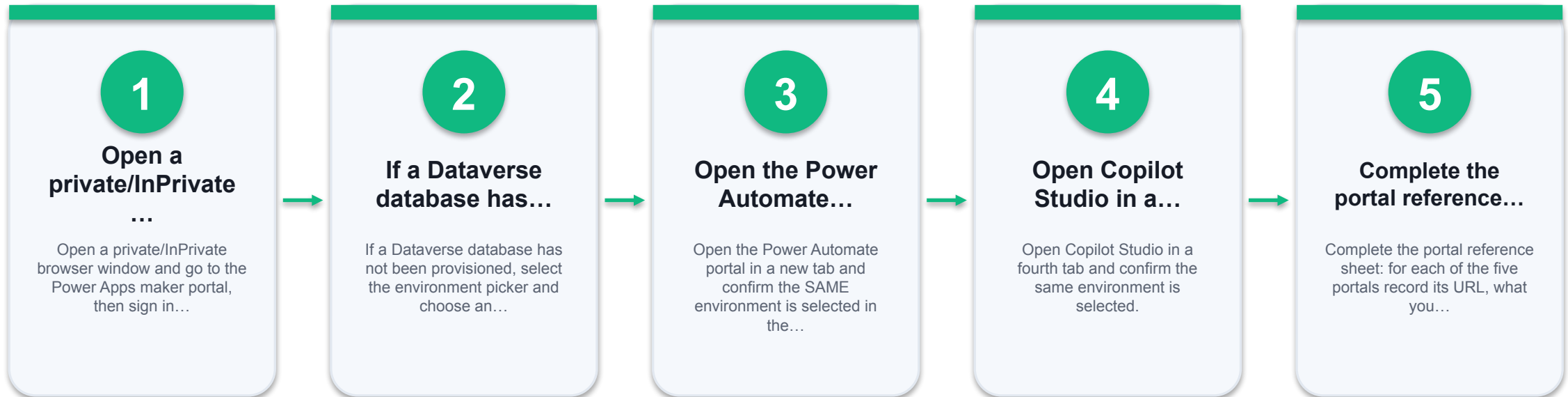
TOOLCHAIN

Power Apps maker portal, Power Automate, Power BI service, Copilot Studio, Power Platform admin center

DONE WHEN

You can open all five portals signed in as the same user, the SAME environment name appears in the picker on the Power Apps, Power Automate and Copilot Studio portals, and Tables shows the standard Dataverse tables.

How Lab 1 Runs



WHY IT MATTERS

Outcome: A validated Power Platform environment plus a portal reference sheet recording your environment name, region, licence type and the Dataverse state.

Validate the Lab Environment and Tour the Maker Portals

✓ Expected result

You can open all five portals signed in as the same user, the SAME environment name appears in the picker on the Power Apps, Power Automate and Copilot Studio portals, and Tables shows the standard Dataverse tables.

IF IT DOESN'T WORK

No environment with Dataverse

Create a free developer environment at <https://aka.ms/PowerAppsDevPlan>, then re-check the environment picker.

'You need permission' on the admin center

You are not an environment or tenant admin. This is expected for a maker account — record what you CAN see and continue.

Wrong environment selected

Every portal keeps its own last-used environment. Set the picker explicitly in each tab before you build anything.

Assess the Business Case and Map Connectors to an Integration Need

LAB 2

You will be able to identify integration opportunities across Contoso's systems and perform a feasibility assessment that selects the right connector for each need.

Contoso's facilities process is spread across email, a spreadsheet and a legacy web system. You analyse the current state, identify where Power Platform can create connections between those systems, and perform a feasibility scan that decides which connector — standard, premium or custom — serves each requirement.

YOU'LL BUILD

A completed integration opportunity register and a connector feasibility assessment recommending a connector and licence tier for each of Contoso's six integration needs.

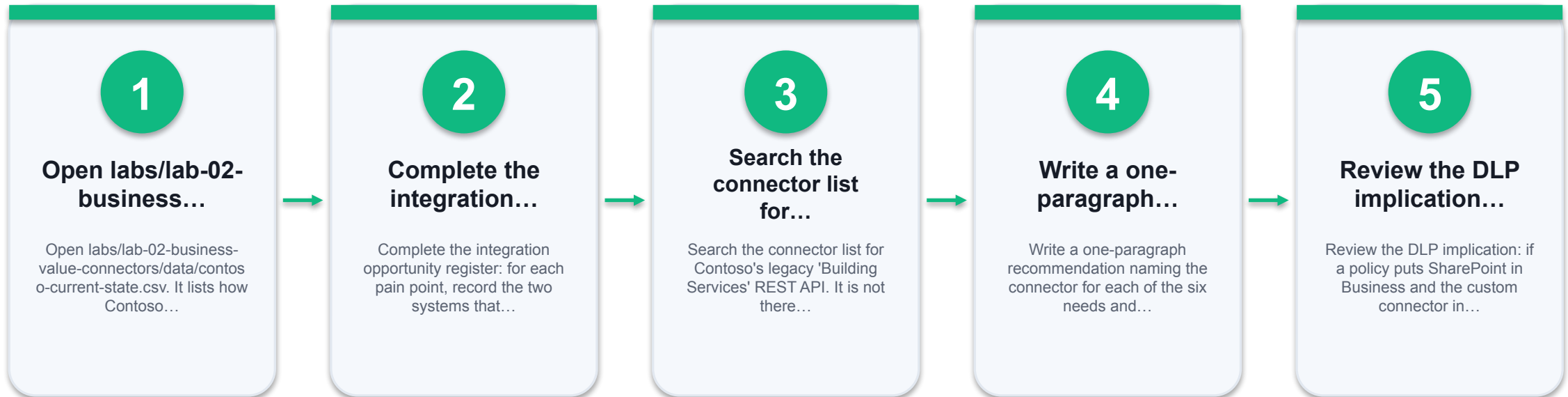
TOOLCHAIN

Power Platform connector reference,
Power Apps maker portal, Microsoft
Learn documentation

DONE WHEN

Your register names both systems and the data direction for every opportunity, and your feasibility assessment recommends a specific connector with its licence tier for all six needs, including a custom connector for the legacy REST API.

How Lab 2 Runs



WHY IT MATTERS

Outcome: A completed integration opportunity register and a connector feasibility assessment recommending a connector and licence tier for each of Contoso's six integration needs.

Assess the Business Case and Map Connectors to an Integration Need

✓ Expected result

Your register names both systems and the data direction for every opportunity, and your feasibility assessment recommends a specific connector with its licence tier for all six needs, including a custom connector for the legacy REST API.

IF IT DOESN'T WORK

Cannot find a connector in the maker portal

Use

<https://learn.microsoft.com/connectors/connector-reference/> — it lists every connector and its tier without needing a licence.

Unsure whether a connector is premium

The connector card shows a PREMIUM badge. Premium requires a per-user or per-app Power Apps/Power Automate licence.

No API for a legacy system

If it exposes no API at all, the integration path is a desktop flow (RPA) driving the UI, not a connector.

Recap — Introduction to Microsoft Power Platform

- Power Platform is five services on a shared data platform and 1,400+ connectors.
- Its business value is speed, agility and delivery capacity — not just licence cost.
- It acts as middleware: it connects, translates, orchestrates and governs between systems (K1).
- Connectors are API wrappers; custom connectors wrap any REST API you own (K5).
- One solution serves web, mobile, desktop and Teams without a rewrite (K3).
- DLP, environments, privacy and accessibility must be designed in from the start.

TOPIC 02

Introduction to Dataverse

Tables, columns, relationships · environments · business logic · administration

Key Concepts — Introduction to Dataverse

1

What Dataverse is

A secure, cloud-hosted business data platform — not just a database. It adds metadata, role-based security, business logic and auditing on top of storage.

2

Beyond a spreadsheet

Enforced data types, relationships, row/column-level security, server-side validation, views, forms and change tracking — none of which a workbook gives you.

3

Tables, columns, rows

Tables define the entity, columns define the fields and their data types, and rows hold the records. Standard, custom and managed (virtual) tables coexist.

4

Relationships

One-to-many, many-to-one and many-to-many relationships connect tables; a lookup column is how a one-to-many relationship appears on a form.

5

Business logic

Business rules (no-code), Power Fx formula columns, real-time workflows, cloud flows and low-code plug-ins — pick the lightest tool that does the job.

6

Environments and ALM

Environments isolate dev, test and production; solutions package the work and Power Platform pipelines move it between them.

THE SHIFT IN THINKING

Dataverse is a platform, not a table.

Storage is the least interesting thing it does — security, metadata, logic and auditing are what make it a business data platform.

WHY NOT JUST USE EXCEL?

Spreadsheet vs Dataverse

Capability	Spreadsheet	Microsoft Dataverse
Data types	A cell holds anything	Enforced per column — text, date, currency, choice, lookup
Relationships	Manual VLOOKUP	First-class 1:N, N:1 and N:N relationships
Security	File-level, all or nothing	Role-based, down to the row and the column
Validation	Optional and easily bypassed	Server-side business rules run for every client
Concurrency	The file locks	Many users write at once, safely
Audit	None	Full change history per row and column
Reuse	One file, one purpose	One model serving apps, flows, BI and agents

WHEN IT MATTERS Contoso's Facilities Log.xlsx is exactly the pain this table describes — file locks, rekeying and no audit trail.

The Anatomy of Dataverse

1

Tables

The entity — Facility Request, Room, Account. Formerly called 'entities'.

2

Columns

The fields and their data types. Formerly called 'attributes' or 'fields'.

3

Rows

The records themselves. Formerly called 'records'.

4

Relationships

How tables connect — one-to-many, many-to-one, many-to-many.

5

Views

A saved, filtered, sorted list of rows — what a grid shows.

6

Forms

The layout used to read or edit a single row.

The Three Types of Table

Type	Where it comes from	Can you change it?	Example
Standard	Ships with Dataverse	Extend it, but not remove it	Account, Contact, User
Custom	You create it	Fully — it is yours	Facility Request, Room
Managed / virtual	Installed by a solution or backed by an external source	Only as the publisher allows	A table surfaced from SQL

WHEN IT MATTERS Build on standard tables where one already fits — Contact already models a person, so do not create 'Employee' from scratch without a reason.

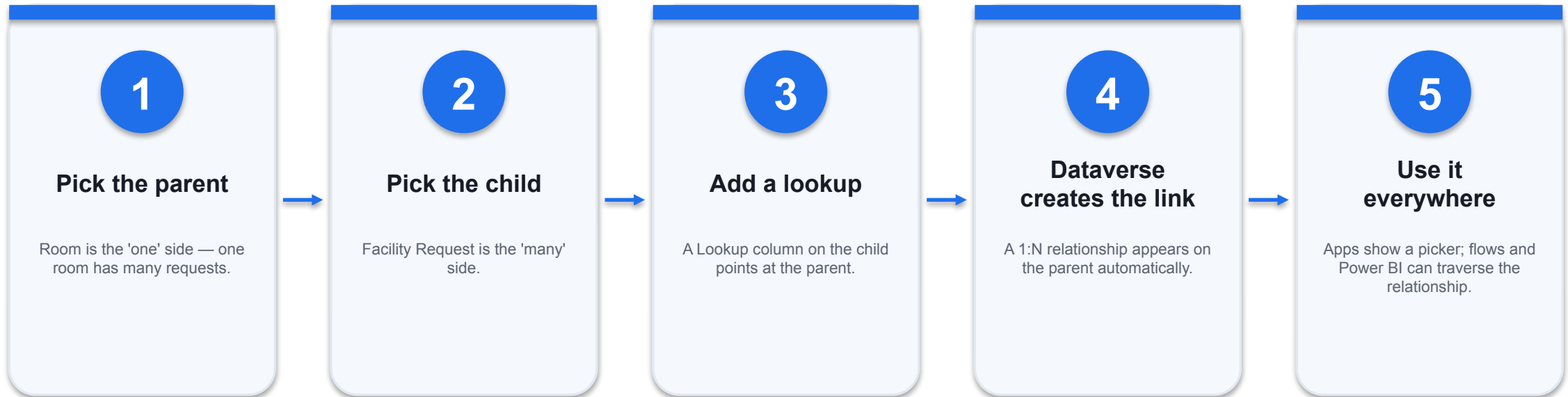
CHOOSE THE RIGHT TYPE FIRST

Column Data Types You Will Use

Data type	Stores	Contoso example
Text / Multiple lines	Characters, up to a max length	Request Title, Description (2000 chars)
Date and time	A date, optionally with time and time-zone behaviour	Date Requested (Date Only, User Local)
Currency	A money value plus a base-currency value	Estimated Cost
Choice	A label AND an integer value	Priority — Low 1, Medium 2, High 3, Urgent 4
Yes/No	A boolean	Conference Room
Lookup	A reference to a row in another table	Room on Facility Request

WHEN IT MATTERS A choice column stores an INTEGER behind the label — Power Automate compares against 3 and 4, not the words 'High' and 'Urgent'. This trips up more people than any other detail.

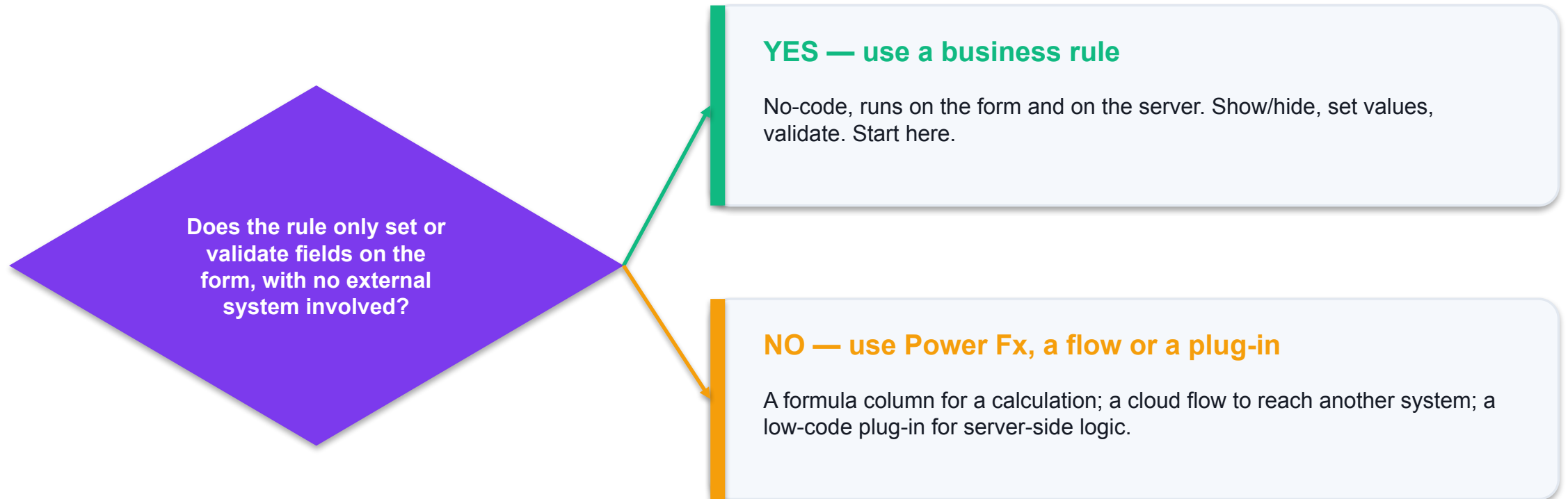
How a Relationship Works



WHY IT MATTERS

The lookup column always lives on the MANY side — this is the single most common exam question about relationships.

Which business logic should I use?



Always choose the lightest tool that does the job — a business rule is cheaper to maintain than a flow, and a flow cheaper than code.

Environments — and why you need more than one

1

Default

Created automatically for the tenant. Everyone is a maker. Never build production here.

2

Developer

Free, single-user, includes Dataverse. Perfect for learning and prototyping.

3

Sandbox

A non-production copy for testing; can be reset or copied from production.

4

Production

The live environment, backed up and governed. Deploy here, do not build here.

5

Isolation

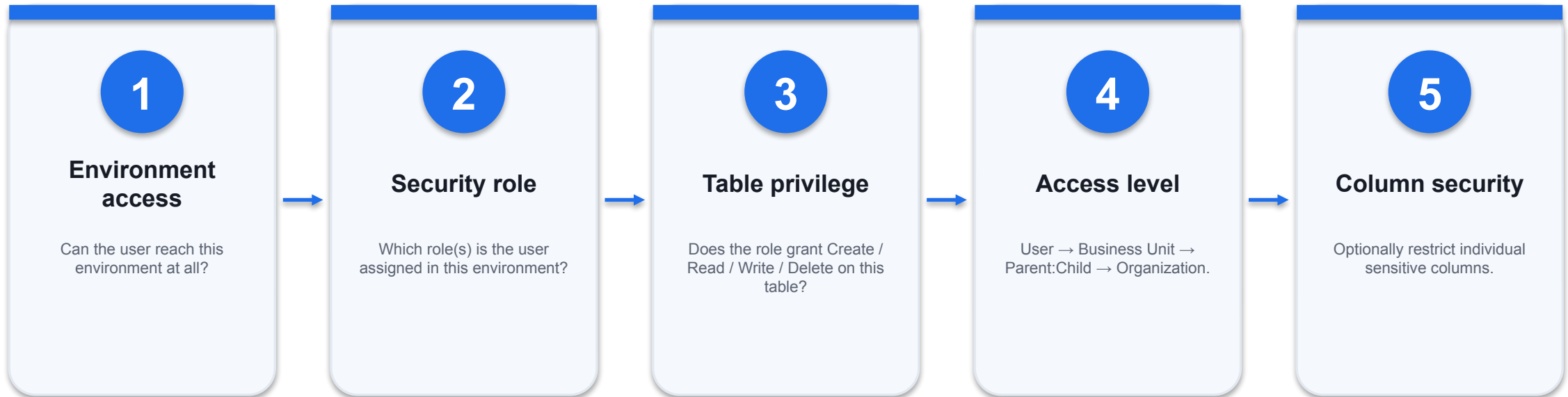
Each environment has its own database, security roles, apps and flows.

6

Region

An environment lives in a region — this drives data residency and latency.

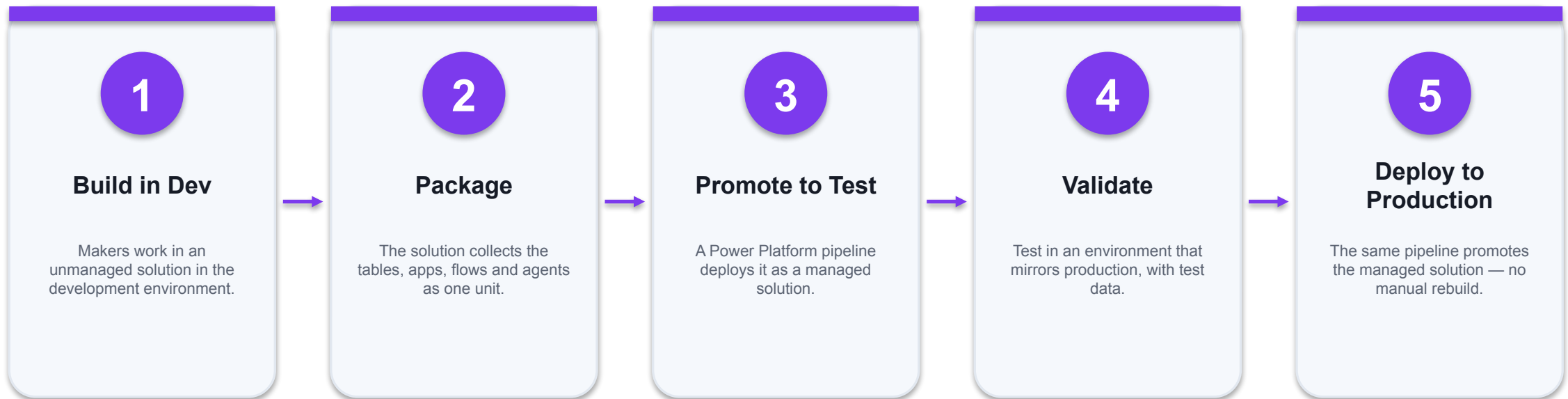
The Dataverse Security Model



WHY IT MATTERS

An employee gets Read at USER level and sees only their own requests; a coordinator gets Read at ORGANIZATION level and sees them all — same table, same role structure.

Application Lifecycle Management (ALM)



WHY IT MATTERS

Unmanaged = the editable source in Dev. Managed = the read-only deployed artefact in Test and Production. Never hand-build the same app twice.

Monitoring, Privacy and Accessibility

1

Admin analytics

Environment, app, flow and Dataverse usage reports in the admin center.

2

Flow run history

Every run, every step, inputs and outputs — the first place to look on failure.

3

Audit history

Dataverse records who changed which column, and when.

4

Data privacy

Tenant-level compliance, retention policies and data residency by region.

5

Accessibility Checker

Built into Power Apps Studio — flags contrast, labels and tab order.

6

Design for everyone

Accessible names on controls, sensible tab order, no colour-only meaning.

Build the Contoso Dataverse Data Model

LAB 3

You will be able to create a Dataverse table with the full range of column types and populate it with business data.

Contoso needs a central store for facilities request data. You create the Facility Request table in Dataverse, add text, date, currency and choice columns, and load the sample request data so later labs have something to display.

YOU'LL BUILD

A Facility Request table in Dataverse with seven columns — including three choice columns — and eight sample request rows.

TOOLCHAIN

Power Apps maker portal, Microsoft Dataverse

DONE WHEN

The Facility Request table lists eight rows; Category, Priority and Status render as choice labels; and Priority values map to Low=1, Medium=2, High=3, Urgent=4.

How Lab 3 Runs



WHY IT MATTERS

Outcome: A Facility Request table in Dataverse with seven columns — including three choice columns — and eight sample request rows.

Build the Contoso Dataverse Data Model

✓ Expected result

The Facility Request table lists eight rows; Category, Priority and Status render as choice labels; and Priority values map to Low=1, Medium=2, High=3, Urgent=4.

IF IT DOESN'T WORK

Table (advanced properties) is missing

You are on the classic Tables view. Use + New table > Table (advanced properties); if unavailable, create the table then rename the primary column afterwards.

Choice values are not 1–4

Open the Priority column, select Edit choices, and correct the Value box next to each label. Lab 8's condition depends on High=3 and Urgent=4.

Currency column will not save

The environment needs a base currency. Open the admin center > Environment > Settings > Product > Currencies and confirm one is set.

Create a Related Table with Copilot and Build a Relationship

LAB 4

You will be able to use AI to create and edit tables and columns, and connect two tables with a lookup relationship.

Contoso's requests need to be tied to a specific room. You create a Room table using the Copilot pane instead of the form designer, then add a lookup column on Facility Request so every request points at one room — a one-to-many relationship.

YOU'LL BUILD

A Room table with five columns and three sample rows, plus a Room lookup column on Facility Request creating a one-to-many relationship.

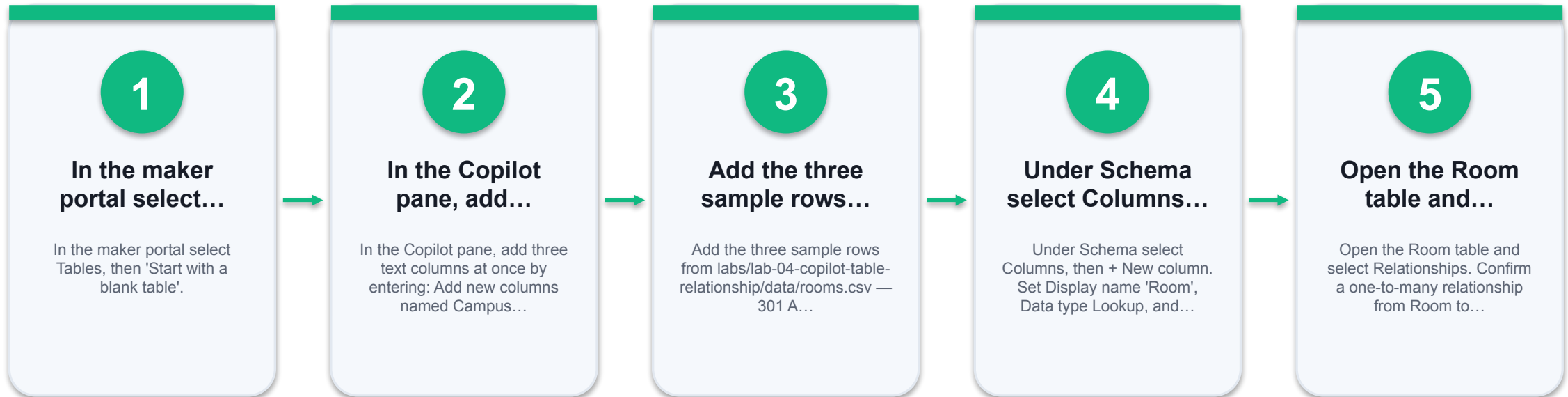
TOOLCHAIN

Power Apps maker portal, Microsoft Dataverse, Copilot in Dataverse

DONE WHEN

The Room table holds three rows, Facility Request has a working Room lookup that resolves to a room name, and the Relationships tab shows the one-to-many link.

How Lab 4 Runs



WHY IT MATTERS

Outcome: A Room table with five columns and three sample rows, plus a Room lookup column on Facility Request creating a one-to-many relationship.

Create a Related Table with Copilot and Build a Relationship

✓ Expected result

The Room table holds three rows, Facility Request has a working Room lookup that resolves to a room name, and the Relationships tab shows the one-to-many link.

IF IT DOESN'T WORK

Copilot pane is not visible

Copilot may be disabled for the environment or region. Add the columns manually with + New column — the learning outcome is the data model, not the tool.

Copilot created the wrong data type

Expected. Open the column, correct the type, and note the correction — reviewing AI output is part of the exam objective.

Lookup shows no rooms

The Room rows were not saved. Reopen the Room table, confirm the rows persisted, then refresh the Facility Request form.

Environments, Security, DLP and Solution-Based ALM

LAB 5

You will be able to describe environments, the security model, data policies and how solutions and pipelines move work between environments.

Contoso must govern what makers can build. You examine the environment list, inspect security roles on the Dataverse table you built, design a DLP policy that separates business from non-business connectors, and package your work into a solution so it can be promoted from development to production.

YOU'LL BUILD

A governance pack: an environment strategy table, a security role mapping for the Facility Request table, a two-group DLP policy design, and an exported unmanaged solution containing your tables.

TOOLCHAIN

Power Platform admin center, Power Apps maker portal, Dataverse security roles, solutions and pipelines

DONE WHEN

You can name the purpose of each environment type, state the row-level scope for both Contoso personas, explain what your DLP policy blocks, and you hold an exported unmanaged solution .zip containing both tables.

How Lab 5 Runs



WHY IT MATTERS

Outcome: A governance pack: an environment strategy table, a security role mapping for the Facility Request table, a two-group DLP policy design, and an exported unmanaged solution containing your tables.

Environments, Security, DLP and Solution-Based ALM

✓ Expected result

You can name the purpose of each environment type, state the row-level scope for both Contoso personas, explain what your DLP policy blocks, and you hold an exported unmanaged solution .zip containing both tables.

IF IT DOESN'T WORK

Data policies is greyed out

DLP policies need tenant or environment admin rights. Complete the design on paper — the exam objective is to DESCRIBE the policy, not to create one.

Solution export fails

A component is still publishing. Select Publish all customizations on the solution, wait for it to finish, then export again.

Cannot see other environments

You only see environments you have a role in. Record what is visible and note why the list is short.

Recap — Introduction to Dataverse

- Dataverse adds metadata, security, logic and auditing on top of storage — it is not a spreadsheet.
- Tables hold columns and rows; relationships connect tables, and the lookup lives on the many side.
- A choice column stores an integer value behind its label — remember Priority 1–4.
- Business rules first, then Power Fx, then flows, then code — always the lightest tool.
- Environments isolate dev, test and production; solutions and pipelines move work between them.
- The security model is layered: environment → role → table privilege → row scope → column.

TOPIC 03

Introduction to Power Apps

Canvas apps · model-driven apps · Copilot-assisted app creation

Key Concepts — Introduction to Power Apps

1

What Power Apps is

A low-code app platform for building business applications that run in a browser and on mobile, over Dataverse or any connector.

2

Canvas apps

Pixel-precise, designer-driven apps. You start from the screen, drag controls onto the canvas and bind them with Power Fx — best for task-focused mobile apps.

3

Model-driven apps

Data-first apps generated from the Dataverse model. Forms, views and navigation come from the schema — best for complex, process-heavy desktop work.

4

Power Fx (K5)

The Excel-like formula language behind canvas apps; functions such as SubmitForm, Navigate, Patch and Filter call the underlying data APIs for you.

5

Copilot-assisted creation

Describe the app in natural language and Copilot drafts the tables, screens and logic — the maker then reviews and refines the result.

6

Plan designer and code apps

Plan designer turns a business problem into a proposed solution set; code apps let pro developers bring their own React/TypeScript UI onto Power Platform.

POWER APPS IN ONE LINE

Two ways to build. One data platform.

Canvas apps start from the screen and give you pixel control. Model-driven apps start from the data and give you the interface for free.

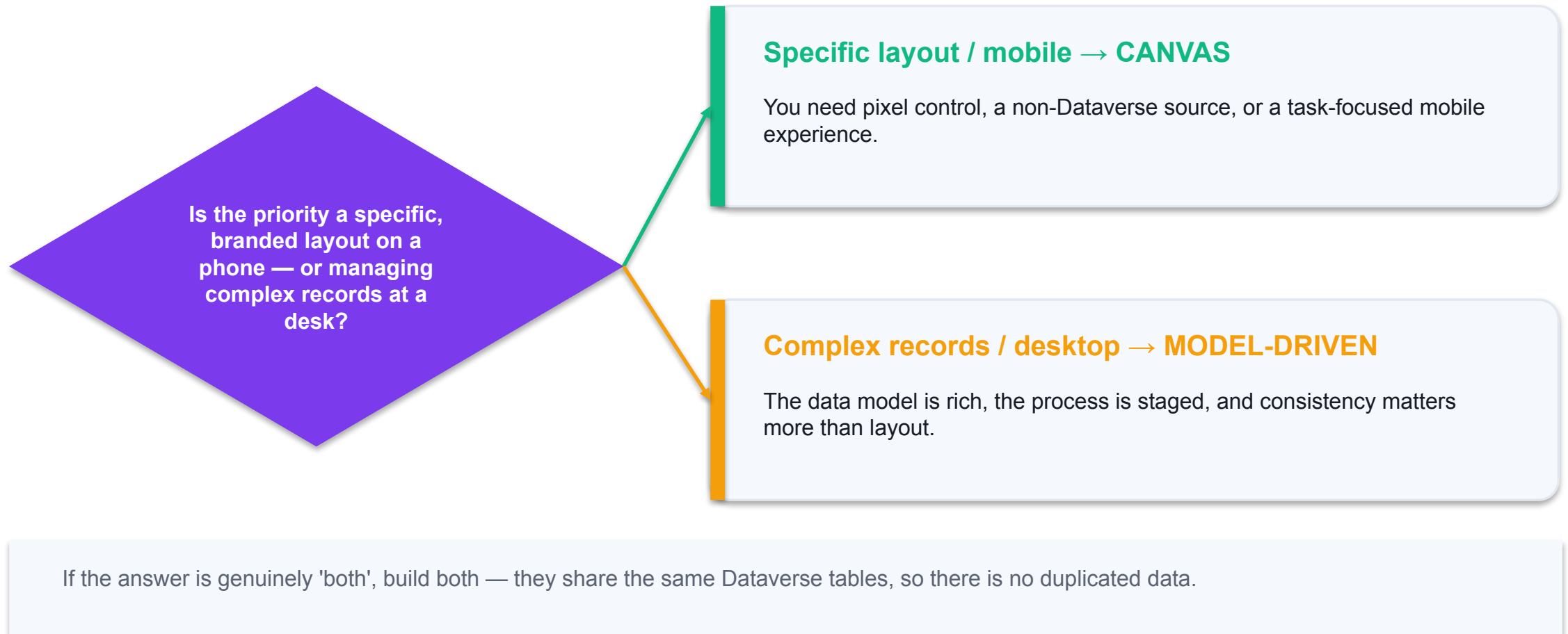
CHOOSE THE RIGHT APP TYPE

Canvas Apps vs Model-Driven Apps

Dimension	Canvas app	Model-driven app
Starting point	A blank screen — you design the layout	The Dataverse data model
UI control	Pixel-precise; you place every control	Generated from forms, views and the site map
Data source	Any connector — Dataverse, SharePoint, SQL, Excel	Dataverse only
Best for	Task-focused mobile and tablet apps	Complex, process-heavy desktop applications
Logic	Power Fx formulas on control properties	Business rules, business process flows, forms
Responsive	You design it, including for phones	Responsive automatically
Contoso use	Employee submits a request from a phone	Facilities team triages all requests

WHEN IT MATTERS This comparison is one of the most heavily weighted areas of the PL-900 exam — be able to pick the right app type from a described scenario.

Canvas or model-driven?



Canvas App Building Blocks

1

Screens

Each screen is a page; apps navigate between them with `Navigate()`.

2

Gallery

A repeating list bound to a data source — the browse screen's core control.

3

Display form

Shows one record's fields, read-only, using data cards.

4

Edit form

Edits or creates one record; `SubmitForm()` writes it back.

5

Input controls

Text input, dropdown, date picker, toggle, slider, attachments.

6

Intelligent controls

Barcode reader, camera, AI Builder components such as text recognition.

Power Fx — the formulas behind the app (K5)

Power Fx is Excel-like and declarative. Each formula sits on a control property and calls the underlying data API for you.

```
// Show the signed-in user in the header
User().FullName

// Reset the form to a NEW blank record, then go to the screen
NewForm(Form2); Navigate('New Room Screen')

// Write the form back to Dataverse, then return to the list
SubmitForm(Form2); Navigate('Facility Requests screen')

// Show only the open, high-priority requests
Filter('Facility Requests', Status = 'Status'.New)
```

User()

Reads the signed-in identity from the Office 365 Users connector — no auth code needed.

NewForm()

Resets the form to create mode. Omit it and SubmitForm UPDATES the last record.

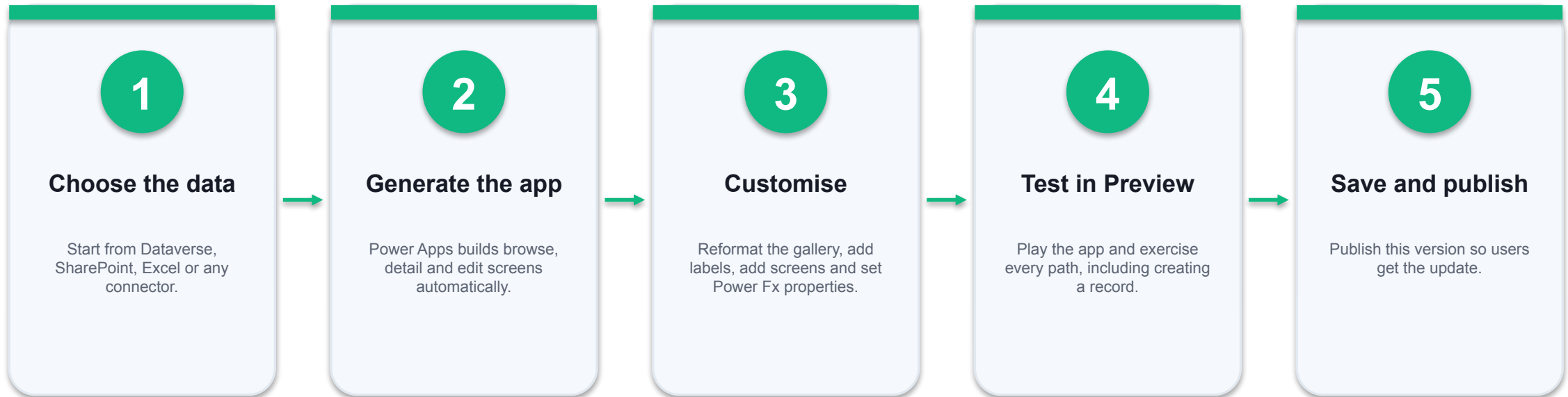
SubmitForm()

Validates the form and writes the row via the Dataverse API — one function, a full API call.

Filter()

Delegates the query to the data source where possible, so large tables stay fast.

Building a Canvas App from Data



WHY IT MATTERS

The generated three-screen app is a starting point, not the finished product — the value is in what you customise afterwards.

Model-Driven App Components

1

Site map

The navigation — areas, groups and subareas the user sees on the left.

2

Forms

Main, quick view, quick create and card forms for a table.

3

Views

Saved queries — Active Facility Requests, My Requests, and so on.

4

Business process flow

Guides the user through stages: Submitted → Approved → In Progress → Done.

5

Business rules

No-code field logic that runs on every client automatically.

6

Charts and dashboards

In-app visualisations built on the same Dataverse views.

AI-Assisted App Creation

1

Describe it in words

Tell Copilot what the app should do and it drafts tables, screens and logic.

2

Copilot in Studio

Ask for changes in natural language while you build — add a screen, change a control.

3

The vibe experience

Iteratively refine an AI-generated app by describing what you want changed.

4

Copilot in Dataverse

Create and edit tables, columns and relationships by prompt — as in Lab 4.

5

Plan designer

Turns a business problem into a proposed set of tables, apps, flows and agents.

6

The maker still owns it

AI drafts; you review, correct and test. Never publish unreviewed output.

Code Apps and Extensibility

1

Code apps

Pro developers bring their own React/TypeScript UI and host it on Power Platform.

2

PCF components

Custom controls that drop into canvas and model-driven apps.

3

When to use them

When the required UX genuinely cannot be built with the standard controls.

4

The trade-off

Full control, but you give up low-code maintainability and need developer skills.

Build a Canvas App on Dataverse

LAB 6

You will be able to build a canvas app connected to a Dataverse table and customise its screens, galleries and forms.

Contoso employees need a mobile-friendly way to see and submit facilities requests. You generate a three-screen canvas app from the Facility Request table, then customise the header with the signed-in user's name and format the gallery so it shows the information a coordinator actually needs.

YOU'LL BUILD

A published canvas app named 'Facility Request App' with a browse gallery, a detail screen, an edit form and a personalised welcome header.

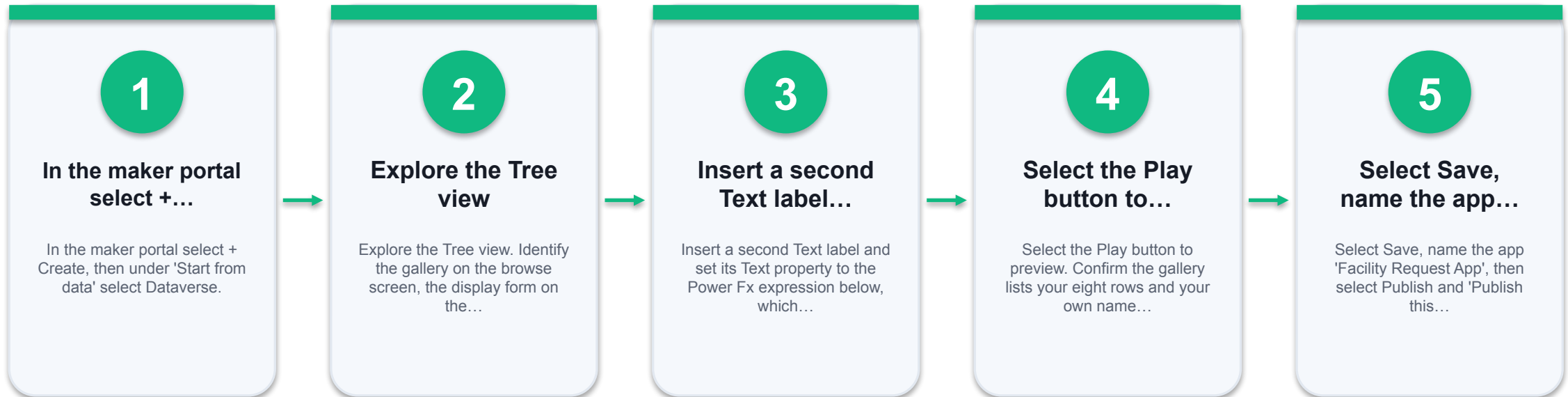
TOOLCHAIN

Power Apps Studio, Microsoft Dataverse, Power Fx

DONE WHEN

The published app opens in Play mode, the gallery lists all eight Facility Request rows, your full name renders in the header, and selecting a row opens its details.

How Lab 6 Runs



WHY IT MATTERS

Outcome: A published canvas app named 'Facility Request App' with a browse gallery, a detail screen, an edit form and a personalised welcome header.

Build a Canvas App on Dataverse

✓ Expected result

The published app opens in Play mode, the gallery lists all eight Facility Request rows, your full name renders in the header, and selecting a row opens its details.

IF IT DOESN'T WORK

User().FullName shows blank

The app needs the Office 365 Users connection. Select Data > + Add data > Office 365 Users, then re-run the preview.

Gallery is empty

The app is pointed at an empty table or the wrong environment. Confirm the environment picker matches Lab 3 and that the table still holds its rows.

Cannot find the Text property

Select the control first, then choose Text in the property dropdown at the top left of the formula bar.

Extend the Canvas App with a New Screen and Power Fx Navigation

LAB 7

You will be able to add a screen and a form to a canvas app and wire navigation with Power Fx functions that call the underlying data API.

Coordinators also need to add rooms without leaving the app. You add a second screen containing an edit form bound to the Room table, then use the Power Fx functions NewForm, SubmitForm and Navigate to move between screens and commit the record — API-level integration expressed as formulas.

YOU'LL BUILD

A second screen, 'New Room Screen', with a working Room edit form and two buttons that navigate between the request list and the room form.

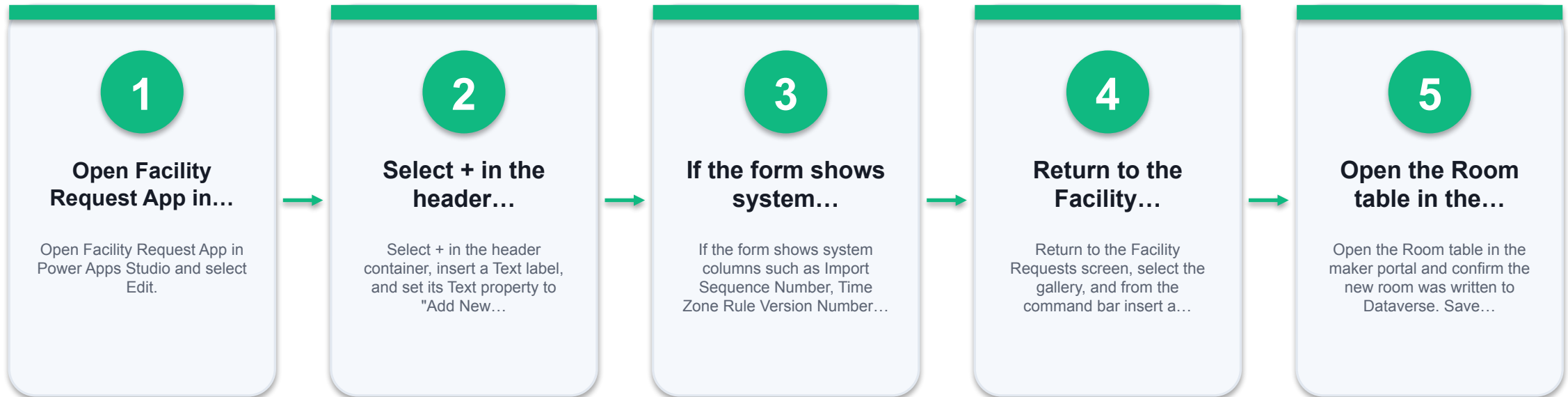
TOOLCHAIN

Power Apps Studio, Microsoft Dataverse, Power Fx

DONE WHEN

Selecting New Room opens a blank Room form; Submit writes the row to the Dataverse Room table and returns you to the request list; the new room is visible in the maker portal.

How Lab 7 Runs



WHY IT MATTERS

Outcome: A second screen, 'New Room Screen', with a working Room edit form and two buttons that navigate between the request list and the room form.

Extend the Canvas App with a New Screen and Power Fx Navigation

✓ Expected result

Selecting New Room opens a blank Room form; Submit writes the row to the Dataverse Room table and returns you to the request list; the new room is visible in the maker portal.

IF IT DOESN'T WORK

Submit does nothing

The formula names the wrong form. Check the form's actual name in Tree view — it may be Form1 rather than Form2 — and update SubmitForm() to match.

The form saves over an existing room

NewForm() is missing from the navigation button. Without it the form keeps the last selected record and SubmitForm updates rather than creates.

Navigate reports the screen does not exist

Screen names are literal. Use the exact name from Tree view, in single quotes if it contains spaces.

Build a Model-Driven App with Custom Forms and Views

LAB 8

You will be able to build a model-driven app from the Dataverse model, customise its form and view, and publish it.

The facilities team manages requests all day on a desktop, so they need a full-featured app rather than a mobile canvas. You build a model-driven app using the modern designer, group its navigation, then customise the Facility Request main form and the Active Facility Requests view so the columns the team needs are visible.

YOU'LL BUILD

A published model-driven app, 'Contoso Facilities Management', with grouped navigation, a customised main form and a customised public view.

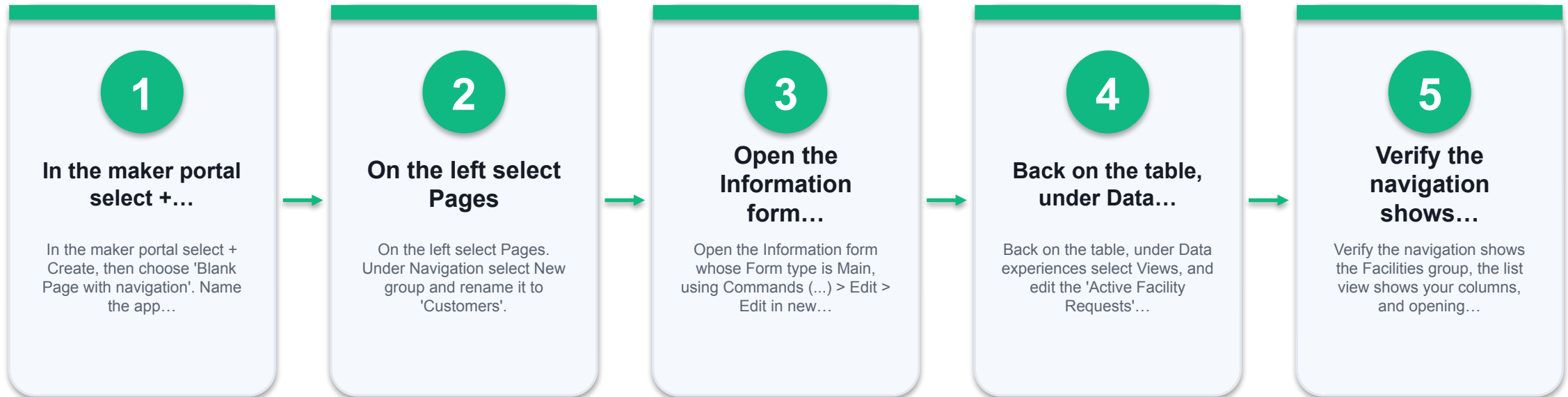
TOOLCHAIN

Power Apps modern app designer,
Microsoft Dataverse, forms and views

DONE WHEN

The published app opens with grouped navigation; the Active Facility Requests view shows the six added columns; and a record's form shows Owner and Status in the header with the other fields in the order you set.

How Lab 8 Runs



WHY IT MATTERS

Outcome: A published model-driven app, 'Contoso Facilities Management', with grouped navigation, a customised main form and a customised public view.

Build a Model-Driven App with Custom Forms and Views

✓ Expected result

The published app opens with grouped navigation; the Active Facility Requests view shows the six added columns; and a record's form shows Owner and Status in the header with the other fields in the order you set.

IF IT DOESN'T WORK

Changes do not appear in the app

Forms and views must be published separately from the app. Select Save and publish in the form/view designer, then Save and Publish the app.

Move down will not place the view in the group

Groups only accept pages below them. Keep selecting Move down until the item indents under the group heading.

The table is missing from + Add page

You are in the wrong environment. Check the environment picker matches Lab 3.

Recap — Introduction to Power Apps

- Canvas apps start from the screen and give pixel control over any data source.
- Model-driven apps start from the Dataverse model and generate the interface for you.
- Galleries browse, display forms read, and edit forms create and update.
- Power Fx functions such as SubmitForm, NewForm, Navigate and Filter call the data API for you.
- Copilot can draft tables, apps and screens — the maker must always review and test the output.
- Plan designer scopes a solution; code apps and PCF extend it when low-code is not enough.

TOPIC 04

Introduction to Power Automate

Cloud, desktop and business process flows · triggers, actions, approvals

Key Concepts — Introduction to Power Automate

1

What Power Automate is

A workflow engine that connects services and automates repetitive work, from an email notification to a full RPA robot driving a legacy desktop app.

2

Three flow types

Cloud flows run in the service; desktop flows (RPA) drive the Windows UI; business process flows guide a person through a staged process.

3

Triggers and actions (K5)

A trigger starts the flow — automated, instant or scheduled — and each action calls a connector operation, which is an API call under the hood.

4

Conditions and loops

Condition, Switch, Apply to each and Do Until add branching and iteration; expressions transform values in transit.

5

Approvals

The Approvals connector sends an actionable request to Teams or Outlook and waits for the outcome before continuing the flow.

6

Copilot in Power Automate

Describe the automation in words and Copilot builds the trigger, condition and actions, which you then verify and test.

POWER AUTOMATE IN ONE LINE

If you do it twice, automate it.

Power Automate connects services and removes the manual steps between them — from one email to a robot driving a legacy application.

The Three Types of Flow

Flow type	What it does	Runs where	Contoso example
Cloud flow	Connects cloud services with triggers and actions	In the Power Platform service	Notify the team on a high-priority request
Desktop flow (RPA)	Drives the Windows/web UI like a person would	On a Windows machine	Read room data from the legacy Building Services app
Business process flow	Guides a PERSON through staged work	Inside a model-driven app	Submitted → Approved → In Progress → Completed

WHEN IT MATTERS A business process flow does not run by itself — it is a guided UI for a human. That distinction is a common exam question.

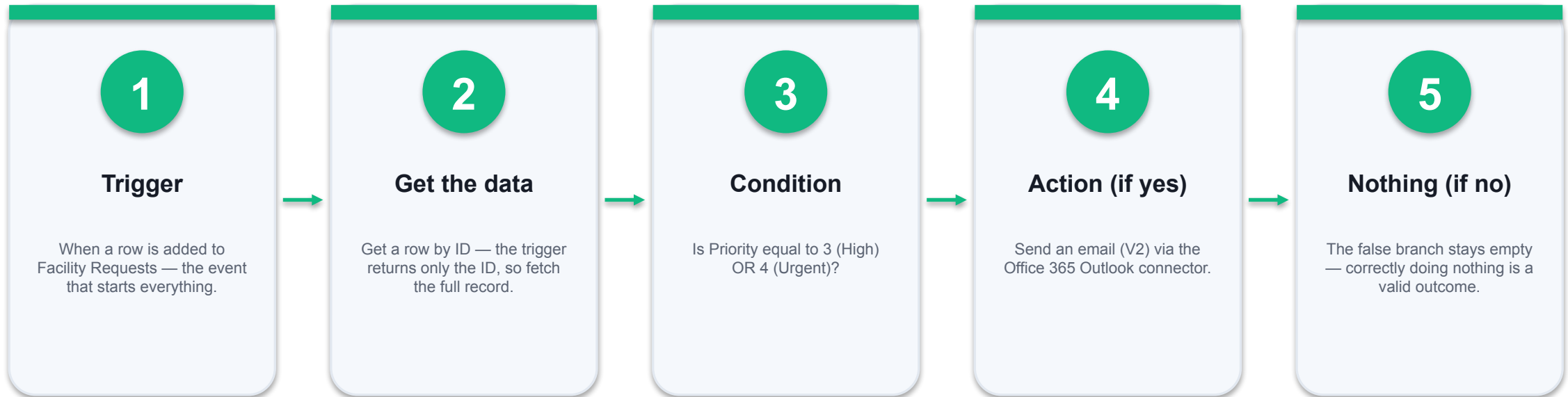
TRIGGERS

The Three Ways a Cloud Flow Starts

Trigger type	Starts when	Example
Automated	An event happens in a connected service	A row is added to Facility Requests
Instant (button)	A person presses a button or runs it manually	'Report an outage' button in Teams
Scheduled	A recurrence timer fires	A weekly open-requests summary every Monday 8am

WHEN IT MATTERS Every cloud flow has exactly ONE trigger and one or more actions. A flow with no trigger cannot exist.

Anatomy of a Cloud Flow (K5)



WHY IT MATTERS

Each action is an API call the connector makes for you — this is API-level integration (A4) without writing a single line of request code.

Control the Flow's Logic

1

Condition

A single if/then/else branch on one or more tests.

2

Switch

Branch on many possible values of one field — cleaner than nested conditions.

3

Apply to each

Loop over every item in an array, running the same actions.

4

Do until

Repeat until a condition becomes true — always set a safety limit.

5

Scope

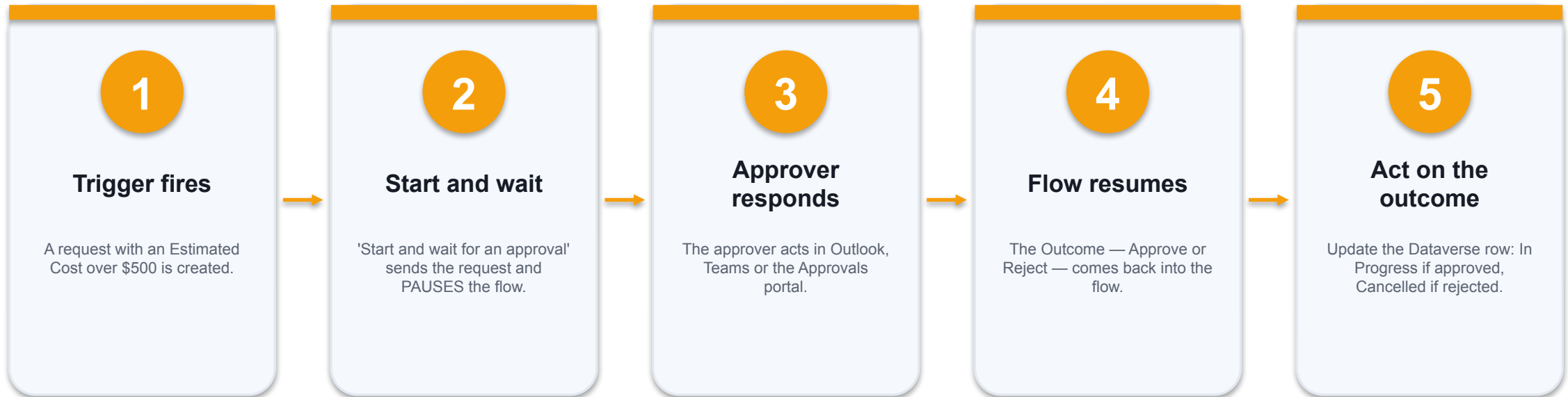
Group actions together so they can be handled or retried as one unit.

6

Expressions

Functions such as `formatDateTime()`, `concat()`, `int()` transform values in transit.

How an Approval Works



WHY IT MATTERS

'Start and wait' genuinely blocks the run until someone responds — a flow that appears stuck at an approval step is usually working exactly as designed.

Where Power Automate Fits in Microsoft 365

1

Outlook

Trigger on mail; send notifications and actionable approval cards.

2

Teams

Post adaptive cards, deliver approvals, and run flows from a message.

3

SharePoint

Trigger on a list item or file; move, copy and set metadata.

4

Forms

Trigger on a survey response and route the answers onward.

5

Document automation

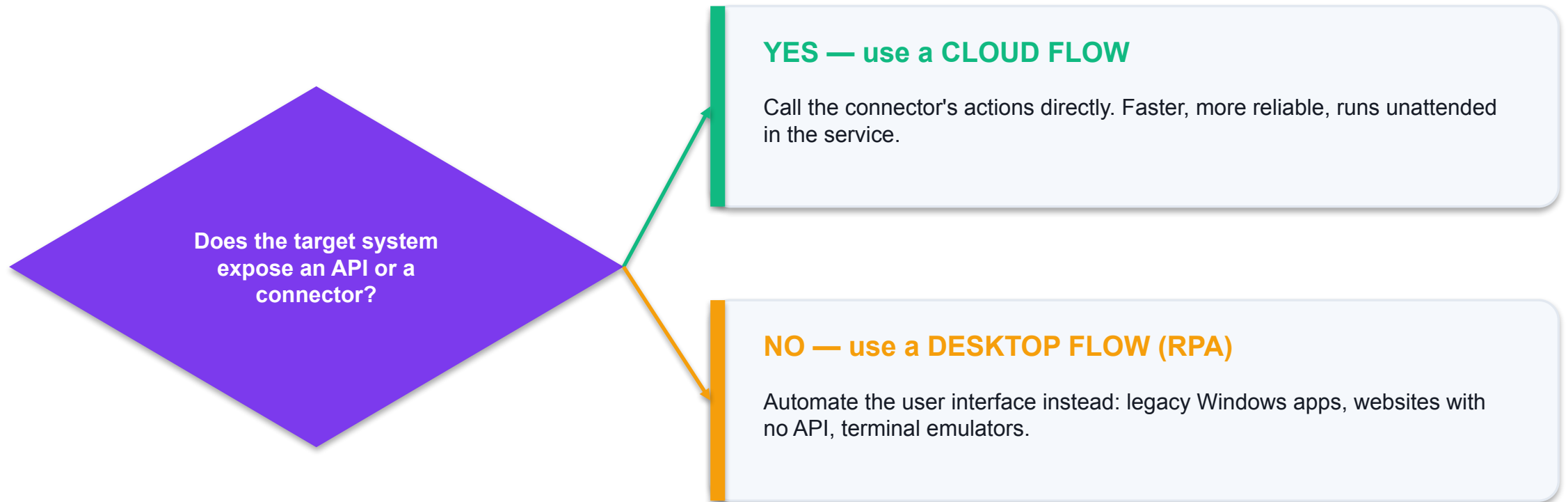
Generate, convert and route documents; AI Builder reads invoices and receipts.

6

Dataverse

Trigger on row added, modified or deleted; create, update and retrieve rows.

Cloud flow or desktop flow?



RPA is the option of last resort — UI automation breaks whenever the interface changes, so prefer an API whenever one exists.

Attended vs Unattended Desktop Flows

Aspect	Attended	Unattended
Who is signed in	A person, at their machine	Nobody — the machine runs it alone
When it runs	While the user works, often triggered by them	On a schedule or from a cloud flow
Best for	Steps that need human judgement or input	High-volume, repeatable back-office batches
Licence	Power Automate Premium	Unattended RPA add-on

Copilot in Power Automate

1

Describe the automation

Write what you want in plain English and Copilot drafts the flow.

2

It builds trigger + actions

Copilot picks the connector, the trigger and the action sequence.

3

Edit by conversation

Ask for a change — add a condition, change the recipient — without the designer.

4

Desktop flows too

Copilot in Power Automate for desktop drafts and explains RPA steps.

5

Always verify

Check the trigger, the condition operators and every dynamic value it chose.

6

Then test it

A flow that saves is not a flow that works — run it and read the history.

Build an Automated Cloud Flow with a Condition

LAB 9

You will be able to create an automated cloud flow triggered by a Dataverse event, add a condition and send a notification through a connector.

Contoso wants the facilities team alerted the moment a high-priority request arrives. You build an automated cloud flow that triggers when a row is added to Facility Request, retrieves the full row, tests the Priority value, and sends an Outlook email only when the request is High or Urgent.

YOU'LL BUILD

A saved cloud flow, 'Notify on High Priority Request', with a Dataverse trigger, a Get a row by ID action, an Or condition on Priority and a Send an email action.

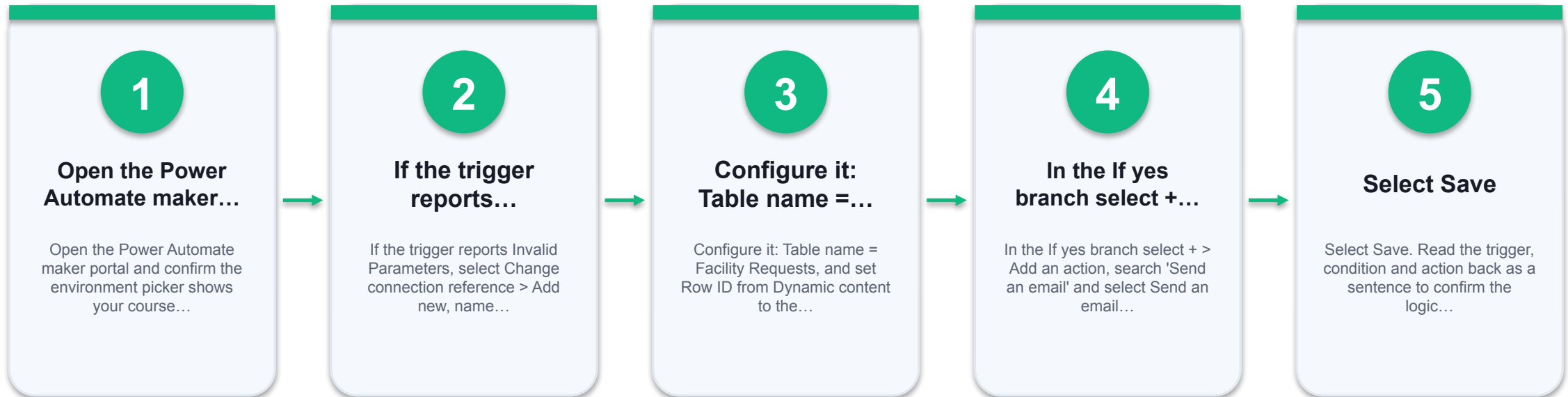
TOOLCHAIN

Power Automate, Microsoft Dataverse connector, Office 365 Outlook connector

DONE WHEN

The flow saves with no validation errors, the condition uses Or across Priority = 3 and Priority = 4, and the Send an email action sits in the If yes branch only.

How Lab 9 Runs



WHY IT MATTERS

Outcome: A saved cloud flow, 'Notify on High Priority Request', with a Dataverse trigger, a Get a row by ID action, an Or condition on Priority and a Send an email action.

Build an Automated Cloud Flow with a Condition

✓ Expected result

The flow saves with no validation errors, the condition uses Or across Priority = 3 and Priority = 4, and the Send an email action sits in the If yes branch only.

IF IT DOESN'T WORK

Trigger shows Invalid Parameters

The Dataverse connection reference is missing. Select Change connection reference > Add new and sign in again.

Priority compares as text and never matches

Use the numeric Value, not the label. High is 3 and Urgent is 4 — the integers you set in Lab 3.

Dynamic content list is empty

The Get a row by ID action has not been saved. Save the flow, reopen it, and the dynamic tokens will appear.

Test the Flow and Diagnose a Failed Run

LAB 10

You will be able to test a flow end to end, read the run history, and diagnose and resolve a failing connection between two application programs.

A flow that saves is not a flow that works. You trigger the flow with real data, read the run history step by step, then deliberately break the integration and diagnose it from the error — the testing and verification skills the C1358 requires.

YOU'LL BUILD

A successful flow run with an email received, plus a fault log recording one induced failure, the error message and the fix applied.

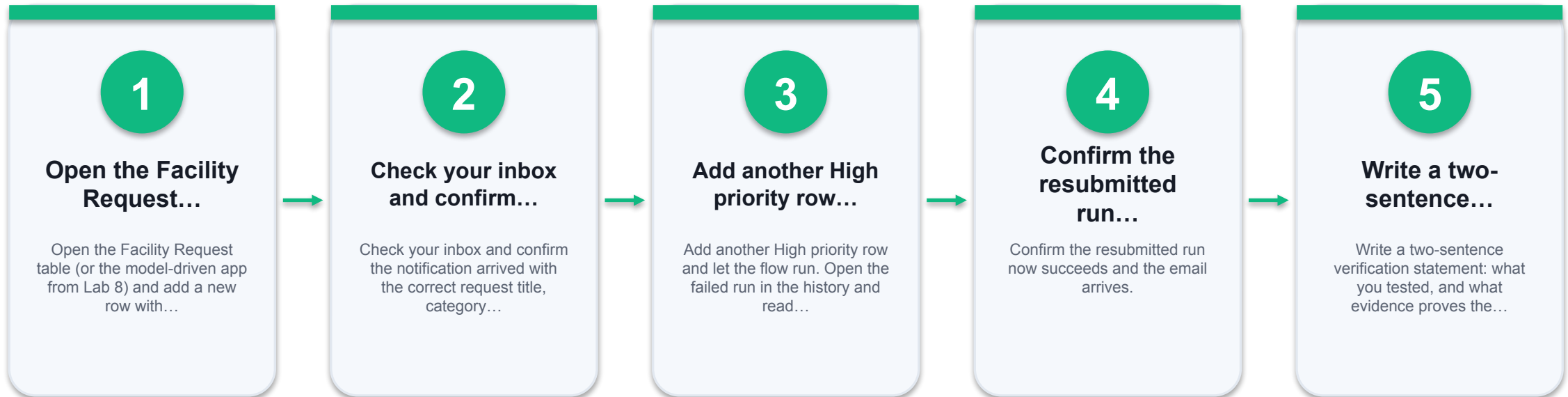
TOOLCHAIN

Power Automate run history, Microsoft Dataverse, Office 365 Outlook

DONE WHEN

Your run history shows at least one Succeeded run that sent an email, one run where the condition correctly evaluated false, and one Failed run that you diagnosed, fixed and successfully resubmitted.

How Lab 10 Runs



WHY IT MATTERS

Outcome: A successful flow run with an email received, plus a fault log recording one induced failure, the error message and the fix applied.

Test the Flow and Diagnose a Failed Run

✓ Expected result

Your run history shows at least one Succeeded run that sent an email, one run where the condition correctly evaluated false, and one Failed run that you diagnosed, fixed and successfully resubmitted.

IF IT DOESN'T WORK

No run appears at all

The trigger scope is User rather than Organization, or the row was added in a different environment. Check both.

Run succeeded but no email arrived

Check the junk folder, then confirm the To address in the action's run inputs is the address you actually monitor.

Resubmit is greyed out

Resubmit only applies to a failed or cancelled run. Open the run itself, not the flow, to find the button.

Approvals, Desktop Flows and Copilot-Assisted Automation

LAB 11

You will be able to describe approvals, Teams/Outlook/SharePoint automation, desktop flows (RPA) and using AI to create and modify flows.

Not every process is a notification. You extend the solution with an approval that pauses the flow until a manager responds, examine where desktop flows fit when a legacy system has no API, and use Copilot to draft a flow from a plain-English description — then review what it produced.

YOU'LL BUILD

An approval flow for high-cost requests, a desktop-flow scenario assessment, and a Copilot-generated flow that you have reviewed and corrected.

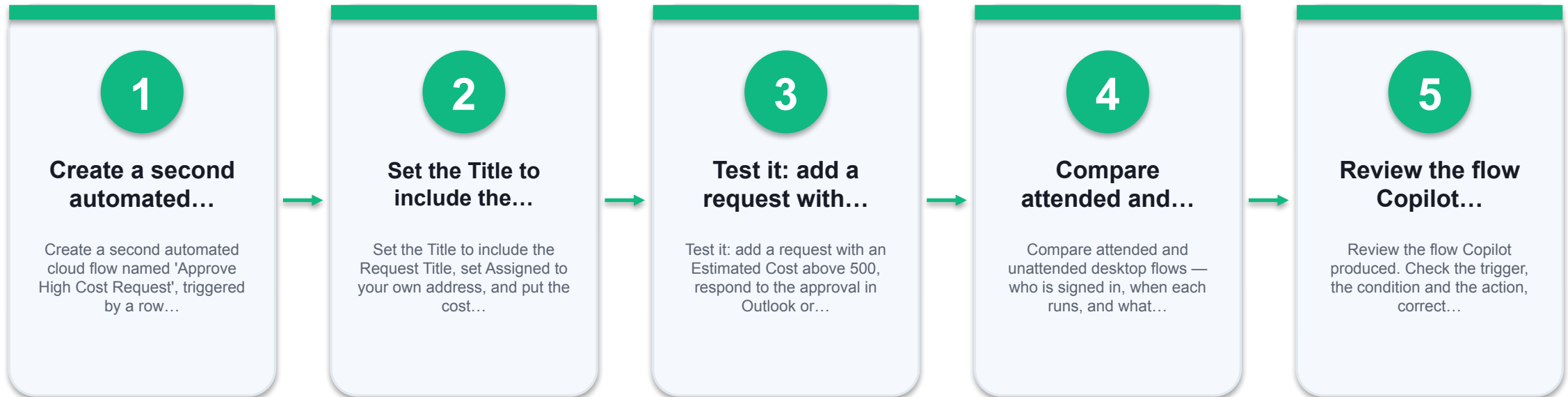
TOOLCHAIN

Power Automate, Approvals connector, Microsoft Teams, Power Automate for desktop, Copilot in Power Automate

DONE WHEN

The approval flow pauses until you respond and then updates the Dataverse row's Status to match your decision; you can state when a desktop flow is required instead of a cloud flow; and you have reviewed and corrected a Copilot-drafted flow.

How Lab 11 Runs



WHY IT MATTERS

Outcome: An approval flow for high-cost requests, a desktop-flow scenario assessment, and a Copilot-generated flow that you have reviewed and corrected.

Approvals, Desktop Flows and Copilot-Assisted Automation

✓ Expected result

The approval flow pauses until you respond and then updates the Dataverse row's Status to match your decision; you can state when a desktop flow is required instead of a cloud flow; and you have reviewed and corrected a Copilot-drafted flow.

IF IT DOESN'T WORK

The approval never arrives

Assigned to must be a full email address on the same tenant. Check the run history inputs for the approval step.

The flow waits forever

'Start and wait for an approval' blocks by design until someone responds. Respond in Outlook, Teams or the Power Automate Approvals page.

Copilot builds the wrong trigger

Expected. Delete the trigger, add the right one manually, and note the correction — reviewing AI output is an exam objective.

Recap — Introduction to Power Automate

- Cloud flows connect services, desktop flows drive a UI, business process flows guide a person.
- Every cloud flow has exactly one trigger — automated, instant or scheduled — and many actions.
- Each action is a connector API call: this is API-level integration without writing request code.
- Conditions, Switch, Apply to each and Do Until add branching and iteration.
- 'Start and wait for an approval' pauses the run until a human responds.
- Use RPA only where no API exists; verify and test everything Copilot generates.

TOPIC 05

Introduction to Power BI

Data modelling · transformation · visualisation · dashboards and sharing

Key Concepts — Introduction to Power BI

1

What Power BI is

The analytics member of the Power Platform: connect to data, model it, visualise it and share the result as reports and dashboards.

2

The parts of Power BI

Power BI Desktop authors the model and report; the Power BI service publishes, shares and refreshes it; the mobile apps consume it.

3

Get and transform

Power Query cleans and reshapes data — remove columns, change types, split, merge and append — before it ever reaches the model.

4

Model and visualise

Relationships join tables into a model; visuals, slicers and filters turn that model into an answerable question.

5

Dashboards vs reports

A report is multi-page and tied to one dataset; a dashboard is a single-page canvas of pinned tiles that can span several reports.

6

Verifying integration (K2, A5, A6)

Refresh history, data lineage and cross-checking totals against the source are how you prove the integration is actually working.

POWER BI IN ONE LINE

Connect. Transform. Model. Visualise.

Power BI is the analytics member of the family — and the place where you prove the integration actually produced correct data.

The Parts of Power BI

1

Power BI Desktop

The free Windows authoring tool — get data, transform, model and design reports.

2

Power BI service

The cloud home — publish, share, refresh, build dashboards and set alerts.

3

Power BI Mobile

Consume reports and dashboards on a phone or tablet, with alerts.

4

Power Query

The transformation engine inside Desktop and the service — clean before you load.

5

Workspaces

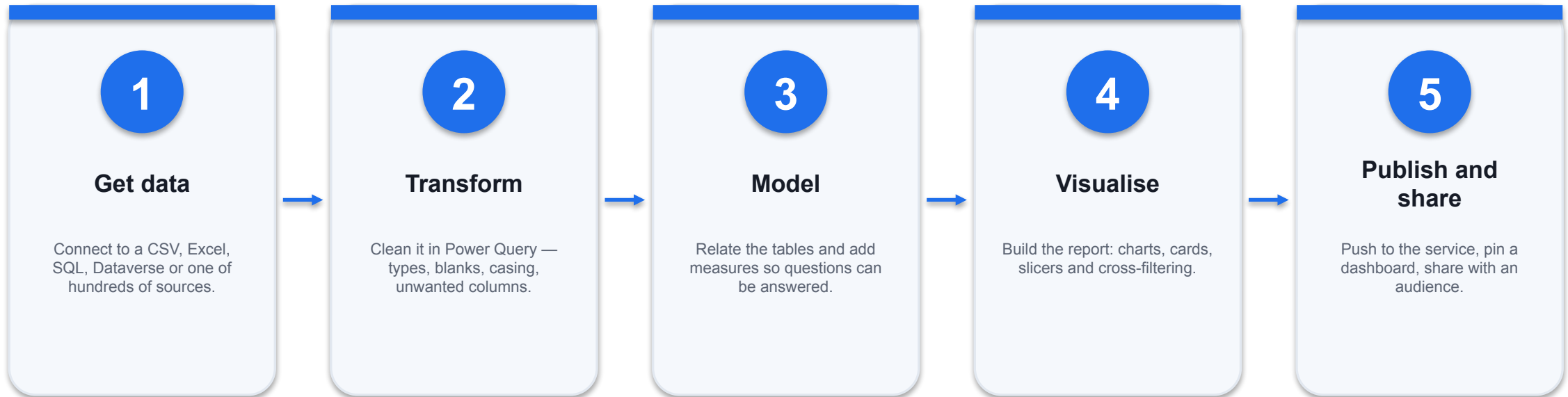
Where content lives and is collaborated on before it reaches an audience.

6

Datasets / semantic models

The reusable model that one or many reports are built on.

The Power BI Workflow



WHY IT MATTERS

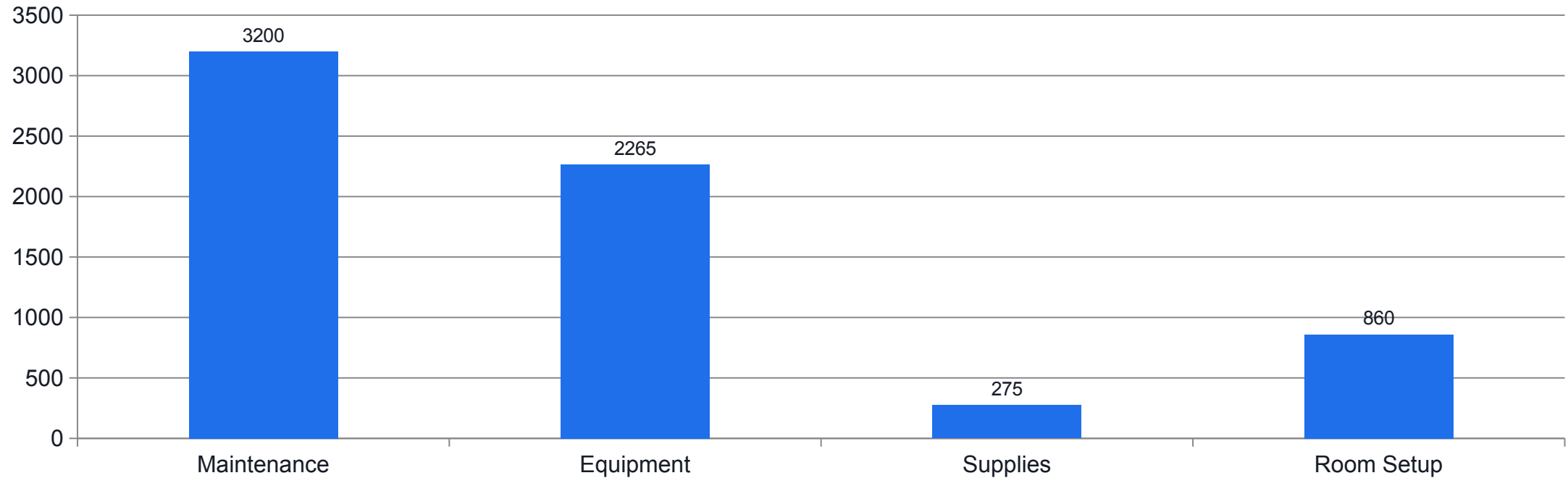
Transform BEFORE you load. Cleaning data after it reaches the model is slower, harder to maintain, and easy to get wrong.

Cleaning Data in Power Query

Problem in the export	Power Query fix	Why it matters
'N/A' in a numeric column	Replace values → null, then Replace null with 0	The column cannot aggregate while it holds text
Blank rows from the export	Remove rows → Remove blank rows	Blank rows inflate counts and skew averages
'equipment' vs 'EQUIPMENT'	Format → Trim, then Capitalize Each Word	Inconsistent casing splits one category into several
An internal ImportBatchId column	Remove columns	Data the report does not need should never reach the model
Dates stored as text	Change type → Date	Text dates cannot be sorted or used in time logic

WHEN IT MATTERS Every one of these faults is present in facility-requests-export.csv. Real exports are never clean — this is the normal state of integration data.

Contoso Facilities Spend by Category



WHAT THE DATA SHOWS

Maintenance dominates spend despite a modest request count — the kind of finding that only becomes visible once the data is modelled, and exactly what management asked for.

Request Volume by Priority



■ Low ■ Medium ■ High ■ Urgent

WHAT THE DATA SHOWS

Half of all requests arrive as High or Urgent — a signal that the priority definitions need tightening, which is a governance finding rather than a technical one.

Report vs Dashboard

Aspect	Report	Dashboard
Pages	Multi-page	Always a single page
Data sources	One semantic model	Tiles pinned from many reports and models
Interactivity	Full — filters, slicers, drill-through	Tiles link back to the source report
Built in	Desktop or the service	The service only
Used for	Exploring and analysing	Monitoring at a glance

WHEN IT MATTERS You cannot pin a visual from Power BI Desktop — pinning only exists in the service, which is why you publish first and pin second.

Verifying the Integration (K2 · A5 · A6)

1

Match the totals

The published card total must equal the figure you recorded in Desktop.

2

Check the row count

Compare the model's row count against the source system's record count.

3

Trace the lineage

The service's lineage view shows source → model → report → dashboard.

4

Read the refresh history

A silently failing refresh is the most common cause of stale figures.

5

Test the relationship

A visual that returns blank for every row means the relationship is wrong.

6

Document the check

Record what you verified and against what — this is the evidence of A6.

Sharing, Security and Administration

1

Workspaces

Collaborate on content before an audience ever sees it.

2

Apps

Package workspace content and distribute it to a wide audience.

3

Sharing a dashboard

The recipient needs access AND an appropriate Power BI licence.

4

Row-level security

Restrict which rows a user sees, defined in the model.

5

Scheduled refresh

Set the cadence the data must be refreshed to stay accurate.

6

Sensitivity labels

Apply Microsoft Purview labels to classify and protect content.

Connect, Transform and Model Facilities Data in Power BI

LAB 12

You will be able to connect Power BI to a data source, clean and transform it with Power Query, and build a model with a relationship.

Contoso's management wants visibility over facilities spend. You connect Power BI Desktop to the facilities export, use Power Query to clean the deliberately messy data, then relate the requests to the rooms so the model can answer questions neither table could answer alone.

YOU'LL BUILD

A Power BI model with a cleaned Facility Requests table, a Rooms table and a working relationship between them.

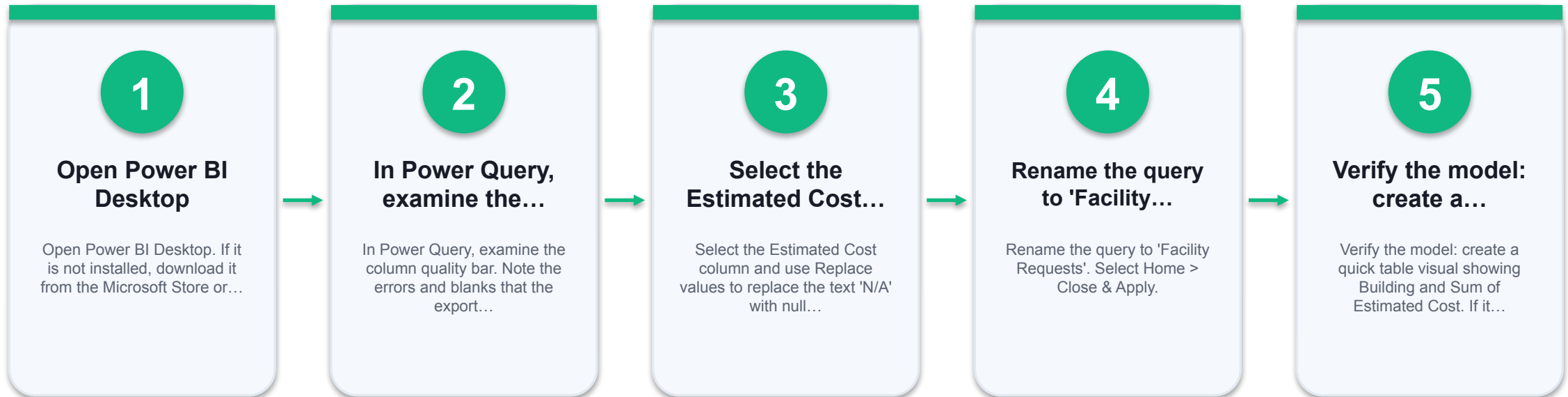
TOOLCHAIN

Power BI Desktop, Power Query Editor

DONE WHEN

Both tables load with no errors, Estimated Cost is numeric with no nulls, Category values are consistently cased, and a table visual of Building against Sum of Estimated Cost returns non-blank figures.

How Lab 12 Runs



WHY IT MATTERS

Outcome: A Power BI model with a cleaned Facility Requests table, a Rooms table and a working relationship between them.

Connect, Transform and Model Facilities Data in Power BI

✓ Expected result

Both tables load with no errors, Estimated Cost is numeric with no nulls, Category values are consistently cased, and a table visual of Building against Sum of Estimated Cost returns non-blank figures.

IF IT DOESN'T WORK

Estimated Cost will not convert to a number

Text such as 'N/A' remains. Replace it with null first, then change the data type.

The relationship will not create

The key columns have different data types or trailing spaces. Trim both and confirm both are Text.

The visual shows blank for every building

The relationship direction is wrong or the key values do not match. Check the room identifiers in both tables are written identically.

Visualise, Publish and Share a Facilities Dashboard

LAB 13

You will be able to build report visuals, publish to the Power BI service, pin a dashboard and share it, then verify the published output against the source.

A model is only useful once people can see it. You build the report visuals management asked for, publish to the Power BI service, pin the key visuals to a dashboard, share it, and then verify the published figures match the source data — the final check that the integration works end to end.

YOU'LL BUILD

A published Power BI report and a shared dashboard showing request volume by category, cost by building and a priority breakdown, with verified totals.

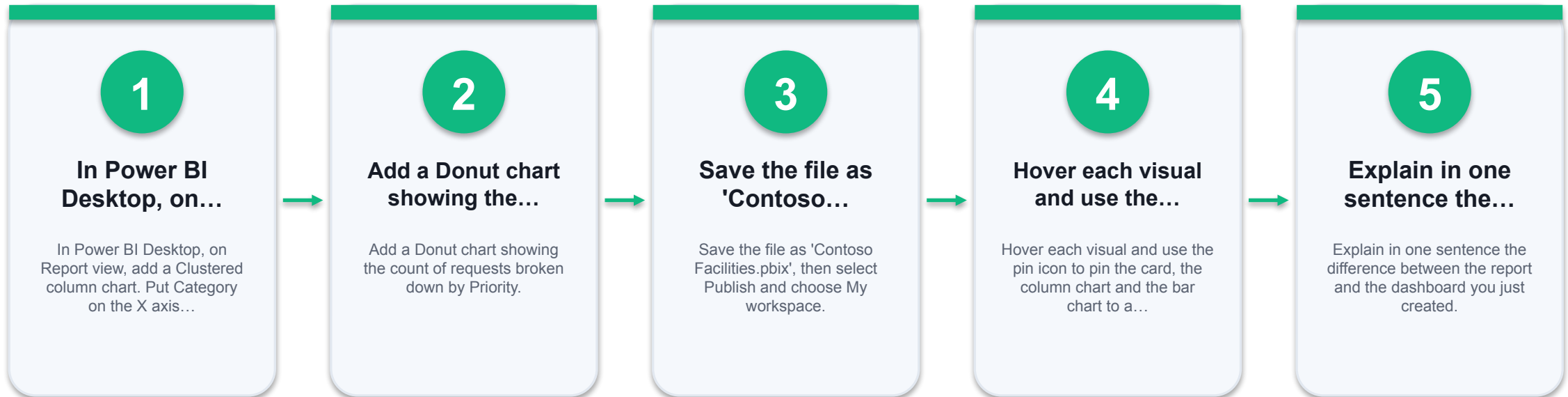
TOOLCHAIN

Power BI Desktop, Power BI service, workspaces, dashboards

DONE WHEN

The report shows five working, cross-filtering visuals; the published report's total Estimated Cost matches the figure recorded in Desktop; and the dashboard holds three pinned tiles that open the report when selected.

How Lab 13 Runs



WHY IT MATTERS

Outcome: A published Power BI report and a shared dashboard showing request volume by category, cost by building and a priority breakdown, with verified totals.

Visualise, Publish and Share a Facilities Dashboard

✓ Expected result

The report shows five working, cross-filtering visuals; the published report's total Estimated Cost matches the figure recorded in Desktop; and the dashboard holds three pinned tiles that open the report when selected.

IF IT DOESN'T WORK

Publish is greyed out

You are not signed in to Power BI Desktop, or the account has no Power BI licence. Sign in at the top right and retry.

Published totals differ from Desktop

A filter is applied on the published page, or the file was changed after publishing. Clear filters and re-publish.

Cannot pin a visual

Pinning is only available in the service, not Desktop. Open the report in app.powerbi.com first.

Recap — Introduction to Power BI

- Power BI Desktop authors, the service publishes and shares, and mobile consumes.
- Power Query cleans and reshapes the data BEFORE it reaches the model.
- Relationships turn separate tables into a model that answers cross-table questions.
- A report is multi-page on one model; a dashboard is one page of tiles pinned in the service.
- Verify published totals against the source — that check is the evidence the integration works.
- Sharing requires both access and a licence; refresh keeps the figures true.

TOPIC 06

Introduction to Power Virtual Agents (Microsoft Copilot Studio)

Agents and topics · knowledge · tools · publishing · troubleshooting integration

Key Concepts — Introduction to Power Virtual Agents (Microsoft Copilot Studio)

1

From Power Virtual Agents to Copilot Studio

Power Virtual Agents is now Microsoft Copilot Studio. The concepts are unchanged — the product builds conversational agents with no code.

2

Agents and topics

A topic is one conversation path: trigger phrases start it and a node tree drives the questions, conditions and messages that follow.

3

Entities and variables

Entities extract meaning from what the user typed; variables carry those values between nodes and into actions.

4

Knowledge sources

Point the agent at SharePoint, a public website or uploaded documents and it answers generatively from that grounded content.

5

Tools and actions

An agent calls a Power Automate flow, a connector or an MCP server to do real work — this is where the agent becomes an integration client (K4).

6

Publishing and troubleshooting

Publish to Teams, a website or a custom channel; then use test chat, the Topic Checker and analytics to find and fix integration failures (A7, A8).

THE NAME CHANGED, THE CONCEPTS DID NOT

Power Virtual Agents is now Copilot Studio.

Same idea, bigger scope: build a conversational agent with no code, ground it in your content, and let it take real action through tools.

Why Organisations Build Agents

1

Deflect repetitive questions

The same facilities questions asked fifty times a week, answered instantly.

2

Available outside office hours

An agent answers at 11pm; the helpdesk does not.

3

Consistent answers

Everyone gets the same policy answer, grounded in the same document.

4

Take action, not just talk

Through tools, the agent creates the request rather than describing how.

5

Escalate cleanly

Hand off to a human with the conversation context attached.

6

Measure and improve

Analytics show which topics work and which need rewriting.

The Components of an Agent

1

Topics

One authored conversation path, started by trigger phrases.

2

Trigger phrases

The example utterances that tell the AI when this topic applies.

3

Entities

Extract meaning from what the user typed — a date, a number, a choice.

4

Variables

Carry values between nodes and into tools and conditions.

5

Knowledge sources

SharePoint, a website or uploaded files the agent answers from generatively.

6

Tools

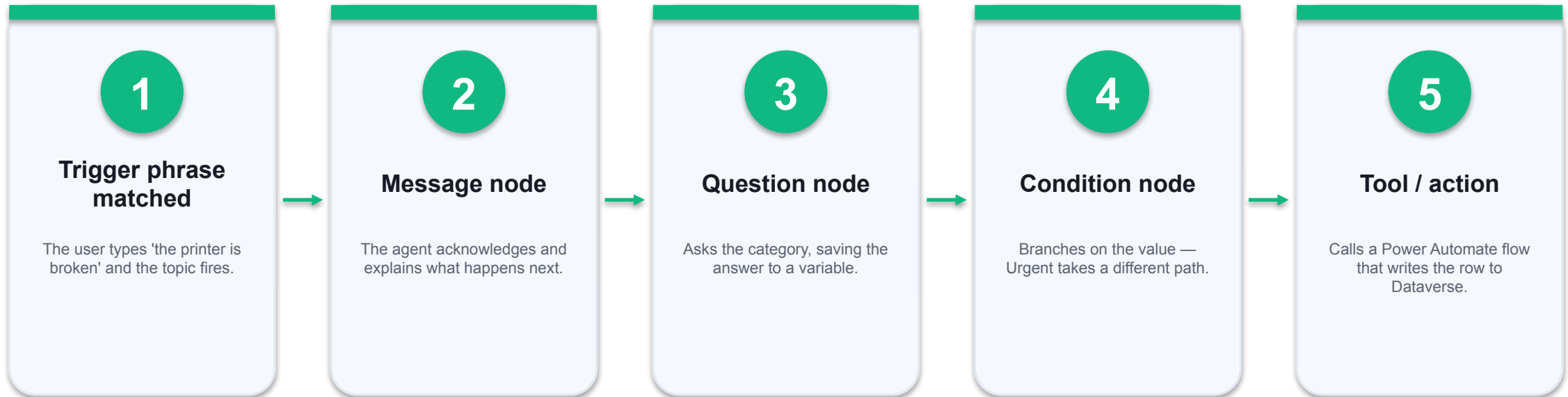
Flows, connectors and MCP servers that let the agent do real work.

Topics vs Knowledge Sources

Aspect	Topic	Knowledge source
How it answers	A path you authored, node by node	Generated from grounded content
Predictability	Deterministic — the same path every time	Varies with the question asked
Best for	Structured tasks: collect details, take an action	Open questions: policy, hours, FAQs
Effort	You design every node	Point at content and let it index
Contoso example	'Report a facility issue' — asks category and priority	'What are the office opening hours?'

WHEN IT MATTERS A good agent uses both: authored topics for anything that must happen the same way every time, and knowledge for the long tail of questions you cannot script.

How a Topic Runs



WHY IT MATTERS

The tool call is the moment the agent stops being a chatbot and becomes an integration client (K4).

Tools — How an Agent Takes Action

1

Power Automate flow

The agent passes variables in and gets results back — as in Lab 15.

2

Connectors

Call any of the 1,400+ connectors directly from a topic.

3

MCP servers

Model Context Protocol servers expose external tools to the agent.

4

Agent flows

Automation authored for and owned by the agent itself.

5

Inputs and outputs

Map topic variables to the tool's inputs; return its outputs to the user.

6

Runs under a connection

The connection's identity needs rights to the target — a common failure.

Publishing to Channels

1

Demo website

A shareable test link — the quickest way to validate with real users.

2

Custom website

Embed the agent in your own site with a code snippet.

3

Microsoft Teams

Where internal employee agents usually belong.

4

Microsoft 365 Copilot

Surface the agent inside the Copilot experience.

5

Other channels

Facebook, Slack and custom channels via Azure Bot Service.

6

Publish every time

Changes are not live until you Publish again — a frequent support call.

Diagnosing Agent Integration Failures (K4 · A7 · A8)

Symptom	Likely root cause	Fix (A8)
The topic never triggers	Trigger phrases overlap another topic	Add distinct phrases; check which topic fired in the test pane
Tool returns an error	A required input is unmapped	Map every input; read the flow run history
Permission denied on the tool	The connection identity lacks table rights	Grant the security role, or use a different connection
Knowledge answers nothing	The source is still processing or generative AI is off	Wait for indexing; enable it in Settings
Users see an old version	The agent was changed but not published	Publish again and hard-refresh the channel

WHEN IT MATTERS This table is the method for Lab 15 and for PP Tasks 4 and 5: observe the symptom, name the root cause, apply the fix, and verify it.

Monitoring and Managing Agents

1

Analytics

Session count, engagement rate, resolution rate and escalation rate.

2

Topic performance

Which topics trigger, which are abandoned, which escalate.

3

Agent evaluations

Test the agent against expected answers to measure quality before release.

4

Microsoft Agent 365

Manage agents as digital workers — identity, access and lifecycle at scale.

5

Test pane and Topic Checker

Catch errors while authoring, before anyone else sees them.

6

Improve the weak topics

Low resolution on a topic is your backlog — rewrite and re-measure.

Build a Copilot Studio Agent with Topics and Knowledge

LAB 14

You will be able to build a conversational agent, author topics with trigger phrases and conditions, and ground it with a knowledge source.

Contoso's facilities inbox is full of the same questions. You build an agent in Copilot Studio, author a topic that collects the request category and priority through a branching conversation, add a knowledge source so the agent answers general questions generatively, and test both in the test chat.

YOU'LL BUILD

A Copilot Studio agent, 'Contoso Facilities Assistant', with a custom topic using a question node and a condition, plus a configured knowledge source.

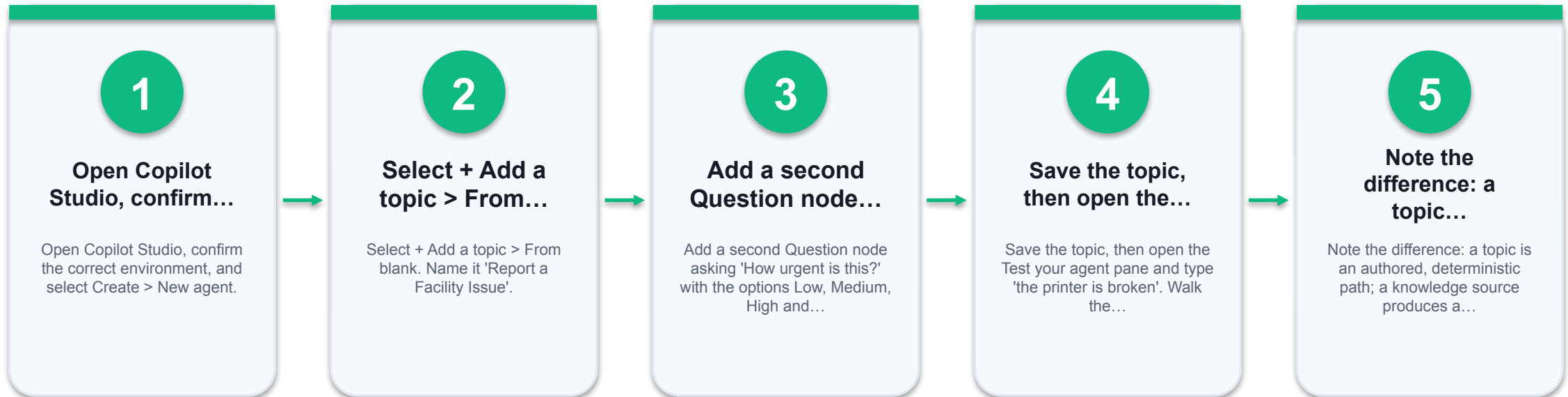
TOOLCHAIN

Microsoft Copilot Studio, topics, entities, variables, knowledge sources

DONE WHEN

Typing a trigger phrase starts your topic, the two questions branch correctly on Urgent versus Low, and a question outside every topic is answered from the knowledge source rather than falling back.

How Lab 14 Runs



WHY IT MATTERS

Outcome: A Copilot Studio agent, 'Contoso Facilities Assistant', with a custom topic using a question node and a condition, plus a configured knowledge source.

Build a Copilot Studio Agent with Topics and Knowledge

✓ Expected result

Typing a trigger phrase starts your topic, the two questions branch correctly on Urgent versus Low, and a question outside every topic is answered from the knowledge source rather than falling back.

IF IT DOESN'T WORK

The topic never triggers

The trigger phrases are too close to another topic's. Add more distinct phrases and re-test; check which topic actually fired in the test pane.

Knowledge answers are not generated

Generative answers may be disabled. Open Settings > Generative AI and confirm the knowledge source is enabled and finished processing.

The condition always takes the false branch

The variable is comparing against the wrong value. Confirm the Question node saves to the variable and that the condition compares the same option.

Add a Tool, Publish the Agent and Resolve Integration Issues

LAB 15

You will be able to call a Power Automate flow from an agent, publish to a channel, then diagnose and resolve integration failures and implement improvements.

An agent that only talks is not integrated. You give the agent a tool — a Power Automate flow that writes the request into Dataverse — publish it to a channel, then work through the integration failures that appear when an agent calls a backend system, log each one and implement the fix.

YOU'LL BUILD

An agent that writes a real Facility Request row to Dataverse through a flow, published to a channel, plus a completed integration fault log.

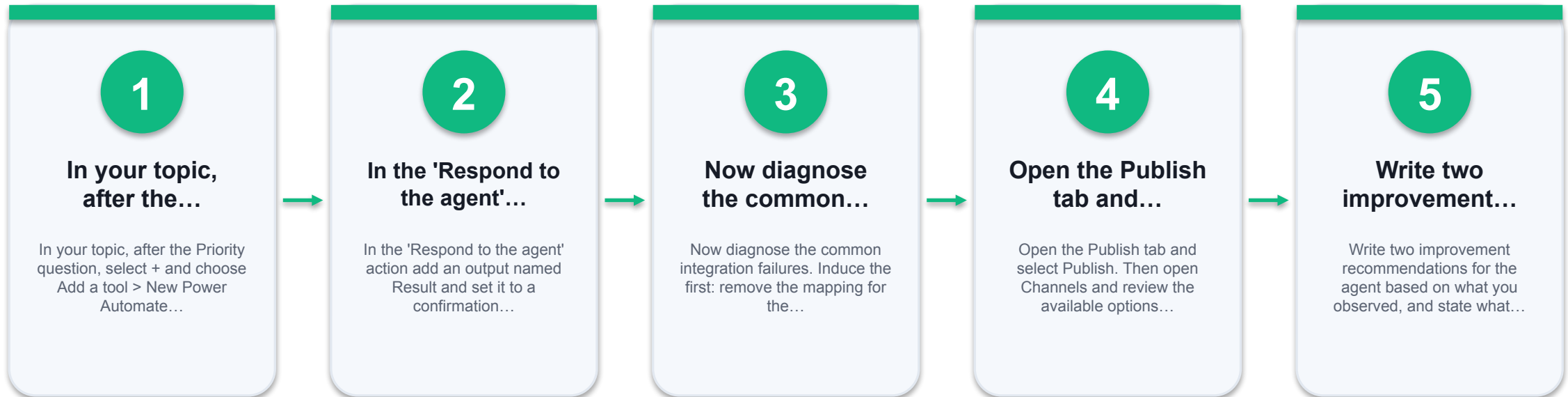
TOOLCHAIN

Microsoft Copilot Studio, Power Automate, Microsoft Dataverse, publishing channels, analytics

DONE WHEN

A completed conversation in the published channel creates a real row in the Facility Request table, and your fault log records at least two induced failures with the error text, root cause and the verified fix for each.

How Lab 15 Runs



WHY IT MATTERS

Outcome: An agent that writes a real Facility Request row to Dataverse through a flow, published to a channel, plus a completed integration fault log.

Add a Tool, Publish the Agent and Resolve Integration Issues

✓ Expected result

A completed conversation in the published channel creates a real row in the Facility Request table, and your fault log records at least two induced failures with the error text, root cause and the verified fix for each.

IF IT DOESN'T WORK

The tool does not appear in Copilot Studio

The flow must be in the same environment and saved. Refresh the tool list, and check the flow's solution is in the agent's environment.

The flow runs but no row is created

A required Dataverse column is unmapped. Open the flow run history and read the Add a new row step's inputs.

Published agent shows an old version

Changes require a fresh Publish. Publish again and hard-refresh the channel link.

Recap — Power Virtual Agents / Copilot Studio

- Power Virtual Agents is now Microsoft Copilot Studio — the concepts are unchanged.
- Topics are authored, deterministic paths started by trigger phrases.
- Knowledge sources answer the long tail generatively from grounded content.
- Tools — flows, connectors, MCP servers and agent flows — let the agent take real action.
- Publish to a channel such as Teams or a website; changes are not live until you publish.
- Diagnose failures from the symptom, fix the root cause, then verify and measure with analytics.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Topic 1 · Business value

Assessed integration opportunities and selected connectors for Contoso.

2

Topic 2 · Dataverse

Built a data model with relationships, and governed it with DLP and solutions.

3

Topic 3 · Power Apps

Built a canvas app and a model-driven app on the same data.

4

Topic 4 · Power Automate

Automated notification and approval, then tested and fixed the flows.

5

Topic 5 · Power BI

Cleaned a messy export, modelled it, and published a verified dashboard.

6

Topic 6 · Copilot Studio

Built, integrated, published and troubleshooted a working agent.

NEXT STEPS

Preparing for the PL-900 Exam

1

Redo the labs

Repeat each lab without the guide until the maker portals feel familiar.

2

Know the comparisons

Canvas vs model-driven, cloud vs desktop flow, report vs dashboard.

3

Know the terminology

Table, column, row, relationship, environment, solution, topic, tool.

4

Practise scenarios

The exam describes a business situation and asks which service fits.

5

Use the Microsoft practice assessment

Free, and closest in style to the real exam.

6

Book the exam

Through Pearson VUE or online proctoring via Microsoft Learn.

Practice Exam

1

1 · Open the practice exam

Go to
<https://exams.tertiaryinfotech.com/practice-exams/microsoft/microsoft-pl-900>

2

2 · Sit it under exam conditions

Time yourself and work without the slides or Learner Guide.

3

3 · Review every explanation

Read the rationale for the questions you got wrong AND right.

4

4 · Revisit the weak topic

Go back to that topic's slides and redo its lab.

5

5 · Re-take it

Repeat until you are consistently comfortable across all six topics.

6

6 · Then book the real exam

Purchase an exam voucher from Tertiary Infotech if you need one.

Recommended Next Courses

1

PL-300
Microsoft Power BI Data Analyst Associate — the deep dive into analytics.

2

PL-200
Microsoft Power Platform Functional Consultant — the next step up from PL-900.

3

AI-102
Microsoft Certified Azure AI Engineer Associate.

4

Copilot & Power Apps
Create Intelligent Power Apps and Power Automate Workflows with Copilot.

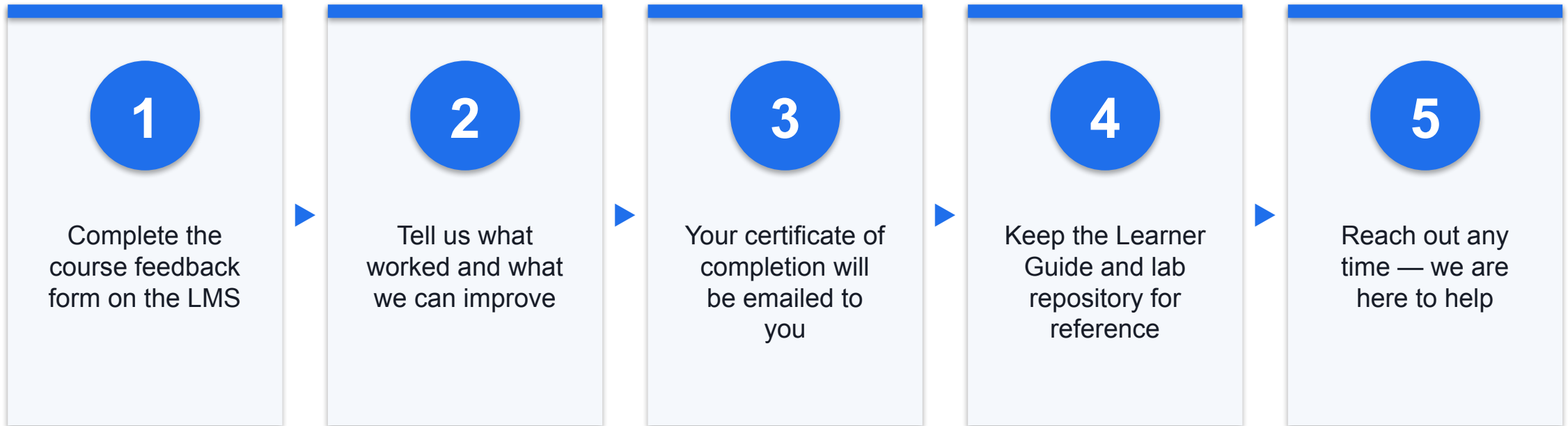
5

AZ-104
Microsoft Certified Azure Administrator Associate.

6

DP-100
Microsoft Azure Data Scientist Associate.

Course Feedback



WE'RE HERE IF YOU NEED US

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LMS / TMS

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5

Lab repository

<https://github.com/tertiarycourses/C1358-Microsoft-Power-Platform-Fundamentals-PL-900>

6

Microsoft Learn

<https://learn.microsoft.com/credentials/certifications/power-platform-fundamentals/>

SEE YOU AT THE NEXT COURSE

Thank You!

You can now identify integration opportunities, build apps, flows, reports and agents on the Power Platform — and you are ready to sit the PL-900 exam.